

Leveraging GxE Interaction to Optimize Multi-Trait Selection in Lima Bean

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Packages

```
library(reshape2)
library(ggplot2)
library(openxlsx)
library(asreml)
```

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```
library(dplyr)
library(tibble)
library(metan)
library(writexl)
```

```
knitr::opts_chunk$set(echo = TRUE)
knitr::opts_knit$set(root.dir = "../")
```

Lima Bean Data

- 3 Locations (Brazil) - Piracicaba - SP, Teresina - PI, Tianguá - CE,
- 40 Lima Bean Lines
- Randomized Complete Block Design (RCBD) - 3 Repetitions
- Plot usable area: 5m²
- Traits:

Trait	Acronym	Unit
Grain Yield	GY	kg/ha
Number days to flowering	NDF	days
Number days to maturity	NDM	days
One Hundred Seed Weight	OHSW	g
Plant Height	PH	cm
Pod Length	PL	mm

Trait	Acronym	Unit
Pod Number	PN	mm
Pod Thickness	PT	mm
Pod Width	PW	mm
Seed Length	SL	mm
Seed Thickness	ST	mm
Seed Width	SW	mm

Load the data

```
LB = read.xlsx("Data//LimaBean.xlsx")
LB$Block    <- as.factor(LB$Block)
LB$Env      <- as.factor(LB$Env)
LB$Genotype <- as.factor(LB$Genotype)
LB$GEN<- as.factor(LB$GEN)
str(LB)
```

'data.frame': 288 obs. of 16 variables:

\$ Env : Factor w/ 3 levels "Piracicaba","Teresina",...: 1 1 1 1 1 1 1 1 1 1 ...

\$ Block : Factor w/ 3 levels "1","2","3": 1 2 3 1 2 3 1 2 3 1 ...

\$ Genotype: Factor w/ 40 levels "H25_53","H25_54",...: 3 3 3 7 7 7 8 8 8 10 ...

\$ GEN : Factor w/ 40 levels "L01","L02","L03",...: 3 3 3 7 7 7 8 8 8 10 ...

\$ NDF : num 42 53 54 59 57 42 42 39 37 59 ...

\$ NDM : num 89 91 92 100 102 89 91 87 107 103 ...

\$ PH : num 50 47 40 48 47 49 46 59 45 52 ...

\$ PL : num 58.4 60 58.2 58.7 60.9 ...

\$ PW : num 13.3 14.2 13.5 13.5 13.4 ...

\$ PT : num 8.53 8.52 8.9 8.3 7.56 8.6 8.11 9.5 8.35 9.5 ...

\$ SL : num 10.8 12 10.6 11 10 ...

\$ SW : num 7.91 7.8 7.74 7.5 7.4 7.03 8.06 8.1 7.8 8.17 ...

\$ ST : num 5.01 5.16 4.84 5.1 5 5 4.71 4.9 4.7 5.2 ...

\$ PN : num 1220 538 615 787 575 ...

\$ GY : num 1125 604 397 945 592 ...

\$ OHSW : num 26.4 26.6 25 25.8 25.3 ...

Histograms

```
hist(LB$NDF)
hist(LB$NDM)
hist(LB$PH)
hist(LB$PL)
hist(LB$PW)
hist(LB$PT)
hist(LB$SL)
hist(LB$SW)
hist(LB$ST)
```

```
hist(LB$PN)
hist(LB$GY)
hist(LB$OHSW)
```

Anova

```
aov = anova_joint(LB, env = Env, gen = GEN, rep = Block, resp = everything())
```

Single environment analyses

```
data.list = split(LB, f = LB$Env)
vccomp=list()
herit=list()
j="Piracicaba"
for (j in names(data.list)) {

  x = droplevels(data.list[[j]])
  cat("====> Environment:", j, fill = TRUE)

  st <- gamem(x,
              gen = GEN,
              rep = Block,
              resp = everything())

  st
  vccomp[[j]]=get_model_data(st, "vcomp")
  vccomp[[j]]$env=j
  herit[[j]]=get_model_data(st, "h2")
  herit[[j]]$env=j
}
```

====> Environment: Piracicaba

Evaluating trait NDF	====	8% 00:00:00
Evaluating trait NDM	=====	17% 00:00:00
Evaluating trait PH	=====	25% 00:00:00
Evaluating trait PL	=====	33% 00:00:00
Evaluating trait PW	=====	42% 00:00:00
Evaluating trait PT	=====	50% 00:00:00
Evaluating trait SL	=====	58% 00:00:00
Evaluating trait SW	=====	67% 00:00:00
Evaluating trait ST	=====	75% 00:00:00
Evaluating trait PN	=====	83% 00:00:00
Evaluating trait GY	=====	92% 00:00:00
Evaluating trait OHSW	=====	100% 00:00:00

P-values for Likelihood Ratio Test of the analyzed traits

model	NDF	NDM	PH	PL	PW	PT	SL	SW	ST	PN
Complete	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Genotype	0.0413	0.683	0.03	0.00124	2.05e-06	0.0766	9.6e-06	9.2e-07	0.77	0.0358

GY OHSW
NA NA
0.113 1.12e-10

Variables with nonsignificant Genotype effect

NDM PT ST GY

====> Environment: Teresina

Evaluating trait NDF	====	8% 00:00:00
Evaluating trait NDM	=====	17% 00:00:00
Evaluating trait PH	=====	25% 00:00:00
Evaluating trait PL	=====	33% 00:00:00
Evaluating trait PW	=====	42% 00:00:00
Evaluating trait PT	=====	50% 00:00:00
Evaluating trait SL	=====	58% 00:00:00
Evaluating trait SW	=====	67% 00:00:00
Evaluating trait ST	=====	75% 00:00:00
Evaluating trait PN	=====	83% 00:00:00
Evaluating trait GY	=====	92% 00:00:00
Evaluating trait OHSW	=====	100% 00:00:00

P-values for Likelihood Ratio Test of the analyzed traits

model	NDF	NDM	PH	PL	PW	PT	SL	SW
Complete	NA	NA	NA	NA	NA	NA	NA	NA
Genotype	8.89e-11	0.00612	1.86e-06	0.0923	2.73e-05	0.36	0.000835	8.9e-06
ST	PN	GY	OHSW					
NA	NA	NA	NA					
0.00292	3.22e-05	2.82e-05	0.000647					

Variables with nonsignificant Genotype effect

PL PT

====> Environment: Tiangua

Evaluating trait NDF	====	8% 00:00:00
Evaluating trait NDM	=====	17% 00:00:00
Evaluating trait PH	=====	25% 00:00:00
Evaluating trait PL	=====	33% 00:00:00
Evaluating trait PW	=====	42% 00:00:00
Evaluating trait PT	=====	50% 00:00:00
Evaluating trait SL	=====	58% 00:00:00
Evaluating trait SW	=====	67% 00:00:00
Evaluating trait ST	=====	75% 00:00:00
Evaluating trait PN	=====	83% 00:00:00

Evaluating trait GY |=====| 92% 00:00:00

Evaluating trait OHSW |=====| 100% 00:00:00

P-values for Likelihood Ratio Test of the analyzed traits

model	NDF	NDM	PH	PL	PW	PT	SL
Complete	NA	NA	NA	NA	NA	NA	NA
Genotype	4.59e-14	0.000249	1.18e-06	0.000304	8.99e-09	0.000596	2.38e-07
SW	ST	PN	GY	OHSW			
NA	NA	NA	NA	NA			
0.000101	0.00037	4e-04	0.00262	5.17e-08			

All variables with significant (p < 0.05) genotype effect

Variance components

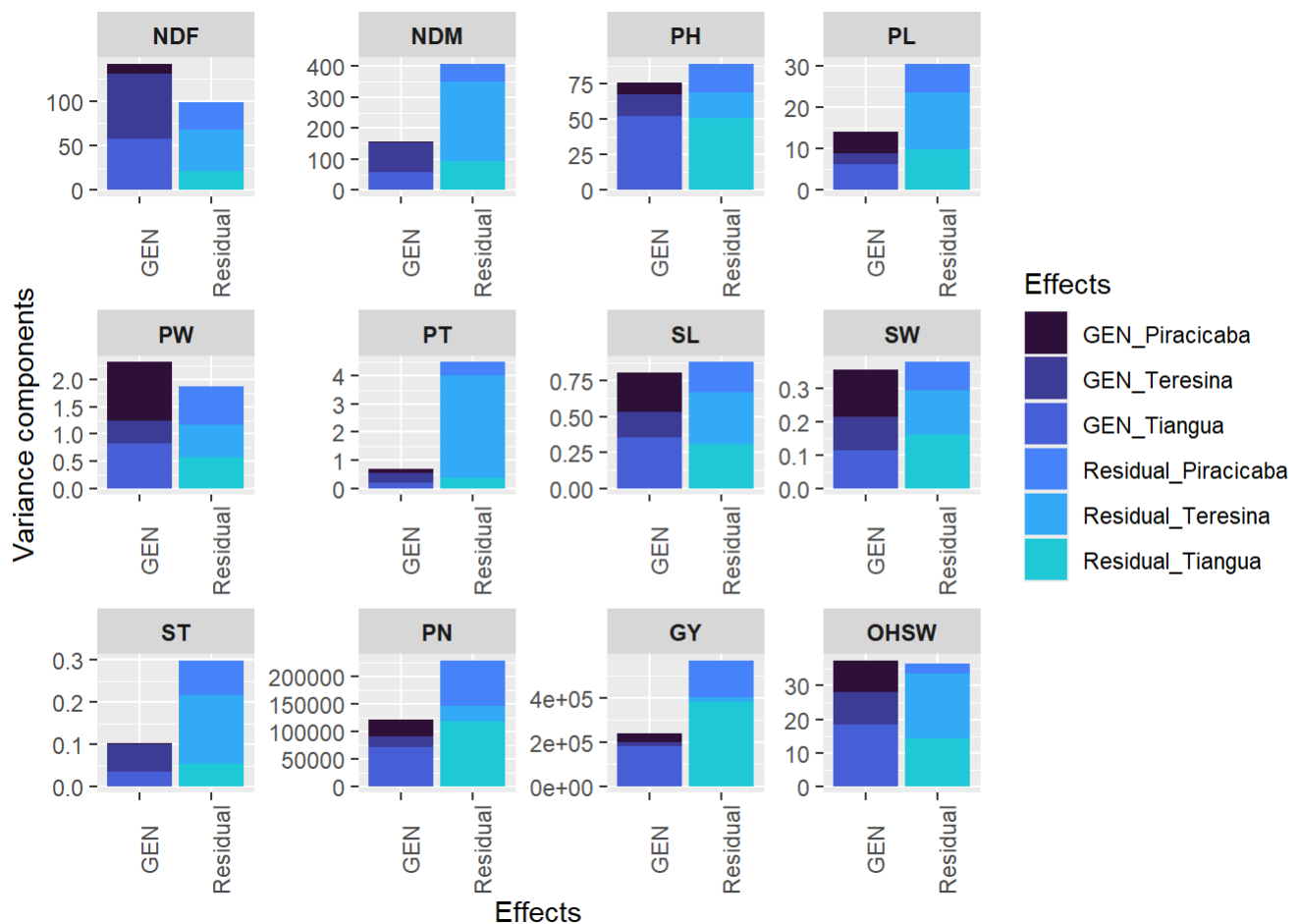
```
custom_palette=viridis::turbo(n = 19)
vc=do.call(rbind,vccomp)
colnames(vc)
```

```
[1] "Group" "NDF"   "NDM"   "PH"    "PL"    "PW"    "PT"    "SL"    "SW"
[10] "ST"    "PN"    "GY"    "OHSW"  "env"
```

```
traits=colnames(vc[,2:13])
vc_long <- reshape2::melt(vc, measure.vars = traits, variable.name = "trait")

vc_long$effect =paste(vc_long$Group,vc_long$env, sep = "_")

vcp <- vc_long |>
  ggplot(aes(x = Group, y = value, fill = effect)) +
  geom_col(position = "stack", just = 0.5) +
  theme(axis.text.x = element_text(angle = 90),
        strip.text = element_text(face = "bold")) +
  facet_wrap(~trait, scales = "free") +
  labs(x = "Effects", y = "Variance components", fill = "Effects") +
  scale_fill_manual(
    values = custom_palette,
  )
vcp
```



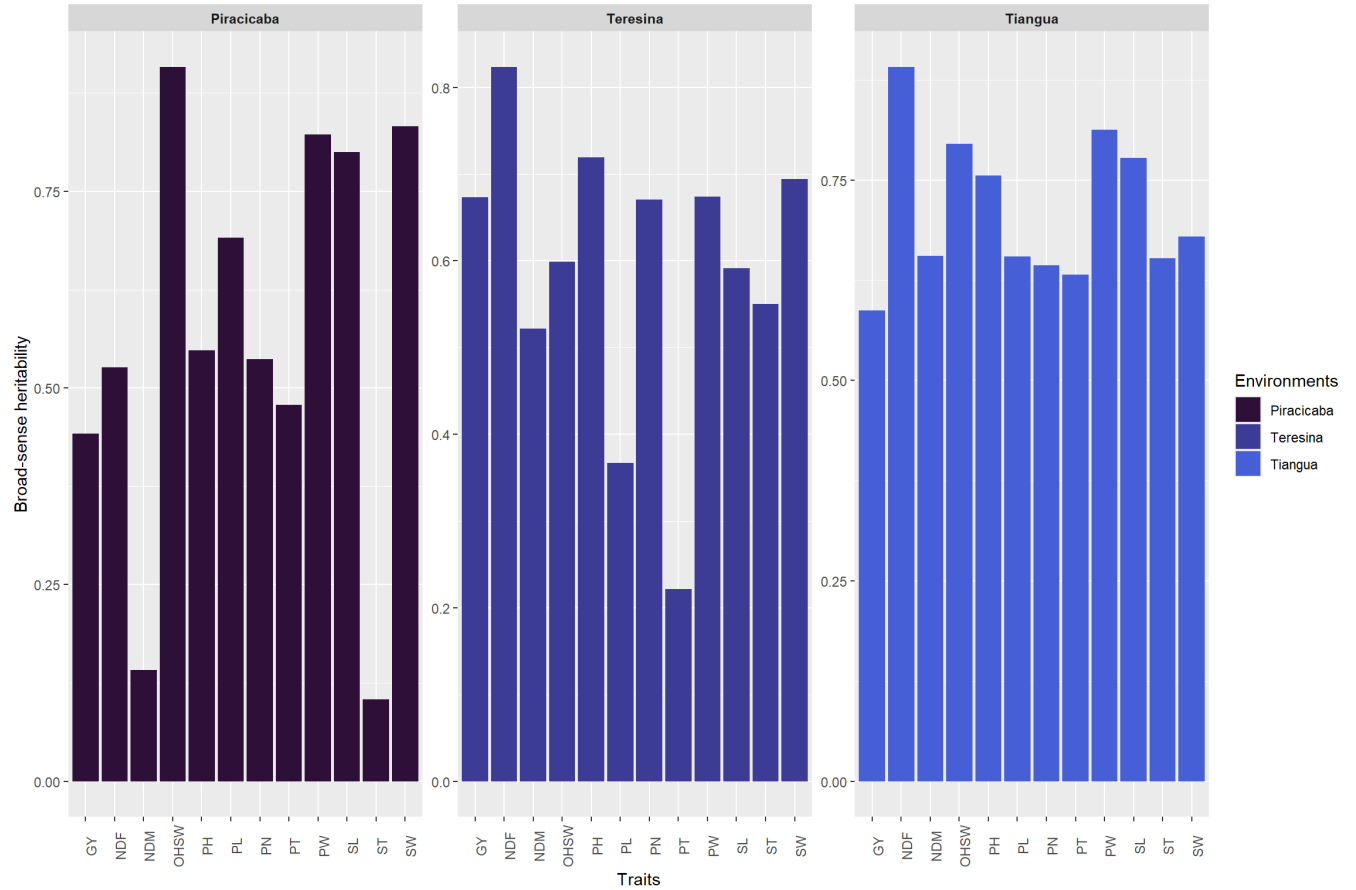
```
ggsave(plot=vcp,device = "pdf",filename = "Plots/varcomp.pdf",dpi = "retina",height = 6,width =
ggsave(plot=vcp,device = "png",filename = "Plots/varcomp.png",dpi = "retina",height = 6,width =
```

Broad-sense Heritability

```
custom_palette=viridis::turbo(n = 19)
her=do.call(rbind,herit)
colnames(her)
```

```
[1] "VAR" "h2"  "env"
```

```
heritp <- her |>
  ggplot(aes(x = VAR, y = h2, fill = env)) +
  geom_col(position = "stack", just = 0.5) +
  theme(axis.text.x = element_text(angle = 90),
        strip.text = element_text(face = "bold")) +
  facet_wrap(~env, scales = "free") +
  labs(x = "Traits", y = "Broad-sense heritability", fill = "Environments") +
  scale_fill_manual(
    values = custom_palette)
heritp
```



```
ggsave(plot=heritp,device = "pdf",filename ="Plots/herit.pdf",dpi = "retina",height = 6,width
ggsave(plot=heritp,device = "png",filename ="Plots/herit.png",dpi = "retina",height = 6,width
```

Mixed-effect models

```
mixedmodel <- gamem_met(LB,env = Env, gen = GEN, rep = Block,random = "gen" ,resp = everything
```

Evaluating trait NDF	====	8% 00:00:00
Evaluating trait NDM	=====	17% 00:00:00
Evaluating trait PH	=====	25% 00:00:01
Evaluating trait PL	=====	33% 00:00:01
Evaluating trait PW	=====	42% 00:00:01
Evaluating trait PT	=====	50% 00:00:02
Evaluating trait SL	=====	58% 00:00:02
Evaluating trait SW	=====	67% 00:00:03
Evaluating trait ST	=====	75% 00:00:03
Evaluating trait PN	=====	83% 00:00:03

Evaluating trait GY |=====| 92% 00:00:04

Evaluating trait OHSW |=====| 100% 00:00:04

P-values for Likelihood Ratio Test of the analyzed traits

model	NDF	NDM	PH	PL	PW	PT	SL	SW
COMPLETE	NA	NA	NA	NA	NA	NA	NA	NA
GEN	3.43e-06	0.0788	4.83e-03	0.34753	3.70e-05	0.00102	8.53e-06	0.000199
GEN:ENV	8.16e-06	0.0120	8.86e-06	0.00107	2.86e-05	1.00000	1.57e-02	0.000918
ST	PN	GY	OHSW					
NA	NA	NA	NA					
0.00083	2.65e-01	0.21557	3.40e-07					
0.46110	5.04e-05	0.00108	3.51e-02					

Variables with nonsignificant GxE interaction

PT ST

```
library(dplyr)
library(tidyr)
library(purrr)

ne <- length(unique(LB$Env))
nr <- length(unique(LB$Block))

varcomp <- map_dfr(names(mixedmodel), function(tr){

  if(!"random" %in% names(mixedmodel[[tr]]))
    return(NULL)
  vc <- mixedmodel[[tr]]$random
  tibble(
    Trait = tr,
    sigma_g = vc$Variance[vc$Group == "GEN"],
    sigma_ge = vc$Variance[vc$Group == "GEN:ENV"],
    sigma_e = vc$Variance[vc$Group == "Residual"]
  )
})

genpar <- get_model_data(mixedmodel) %>%
  pivot_longer(
    cols = -Parameters,
    names_to = "Trait",
    values_to = "Value"
  ) %>%
  pivot_wider(
    names_from = Parameters,
    values_from = Value
  )

herit_table <- varcomp %>%
  left_join(genpar, by = "Trait") %>%
  dplyr::mutate(

    H2_individual =
      sigma_g /
```



```

(sigma_g + sigma_ge + sigma_e),

H2_MET =
  sigma_g /
  (sigma_g +
    sigma_ge/ne +
    sigma_e/(ne*nr)),

H2mg_metan = h2mg) %>%
dplyr::select(
  Trait,
  sigma_g,
  sigma_ge,
  sigma_e,
  H2_individual,
  H2_MET,
  H2mg_metan,
  Accuracy,
  GEIr2,
  rge,
  CVg,
  CVr,
  `CV ratio`)
herit_table

```

A tibble: 12 × 13

	Trait	sigma_g	sigma_ge	sigma_e	H2_individual	H2_MET	H2mg_metan	Accuracy
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	NDF	38.9	16.9	3.41e+1	0.432	0.805	0.805	0.897
2	NDM	25.5	35.7	1.54e+2	0.119	0.468	0.468	0.684
3	PH	12.1	14.5	3.02e+1	0.213	0.597	0.597	0.772
4	PL	0.947	3.51	1.08e+1	0.0620	0.286	0.286	0.534
5	PW	0.423	0.273	6.15e-1	0.322	0.726	0.726	0.852
6	PT	0.728	0	1.57e+0	0.316	0.806	0.806	0.898
7	SL	0.199	0.0676	3.09e-1	0.346	0.778	0.778	0.882
8	SW	0.0727	0.0430	1.32e-1	0.294	0.715	0.715	0.846
9	ST	0.0376	0.00639	1.06e-1	0.250	0.729	0.729	0.854
10	PN	9271.	31235.	7.21e+4	0.0824	0.335	0.335	0.579
11	GY	22388.	58803.	1.81e+5	0.0853	0.360	0.360	0.600
12	OHSW	10.0	2.56	1.38e+1	0.379	0.807	0.807	0.898

i 5 more variables: GEIr2 <dbl>, rge <dbl>, CVg <dbl>, CVr <dbl>,
`CV ratio` <dbl>

Broad-sense Heritability MET

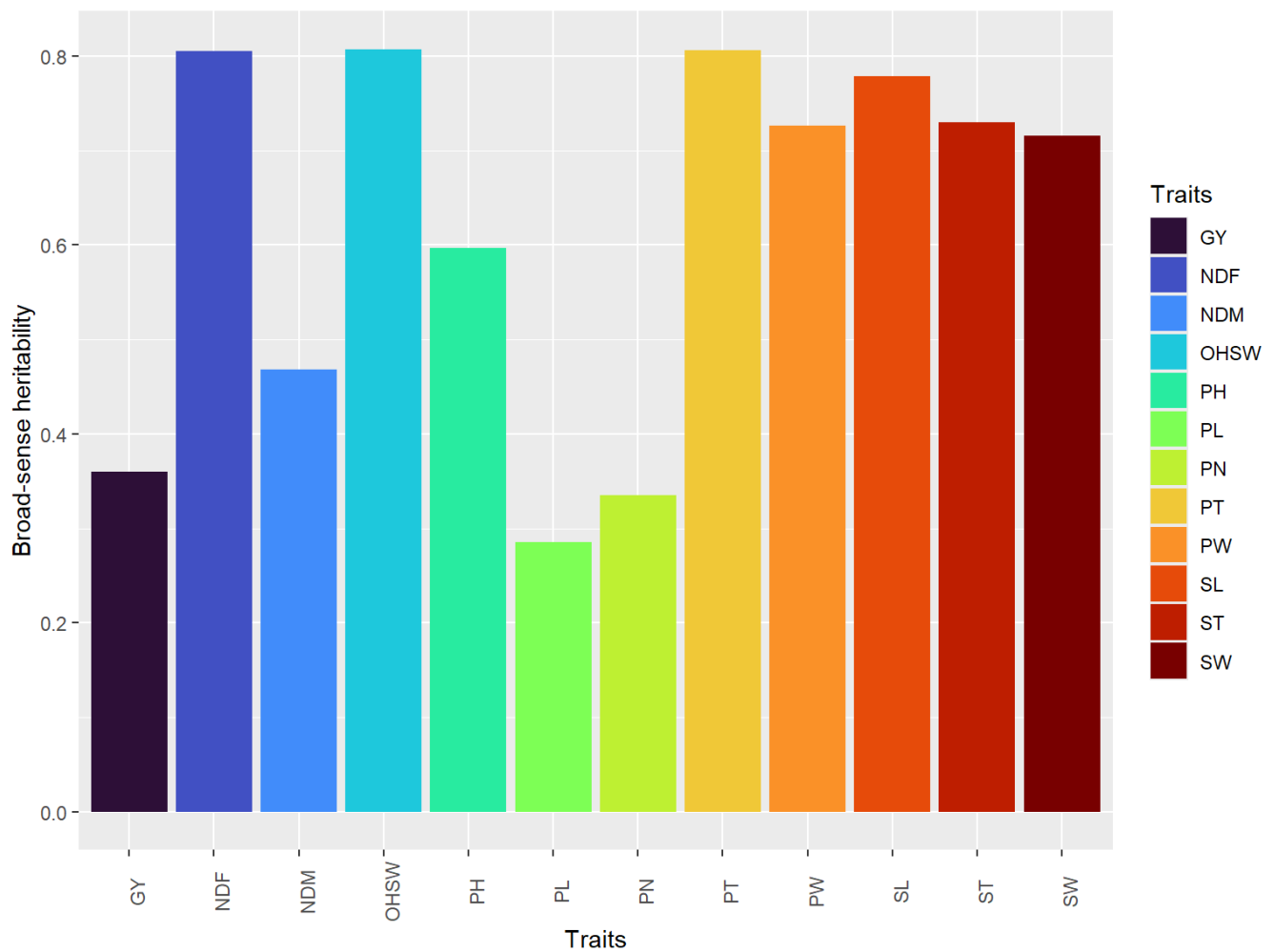
```

herit_met=get_model_data(mixedmodel, "h2")

custom_palette=viridis::turbo(n = 12)
heritp2 <- herit_met |>
  ggplot(aes(x = VAR, y = h2, fill=VAR)) +
  geom_col(position = "stack", just = 0.5) +
  theme(axis.text.x = element_text(angle = 90),
        strip.text = element_text(face = "bold")) +

```

```
labs(x = "Traits", y = "Broad-sense heritability", fill = "Traits") +
scale_fill_manual(
  values = custom_palette,
)
heritp2
```



```
ggsave(plot=heritp2, device = "pdf", filename = "Plots/herit_met.pdf", dpi = "retina", height = 6, width = 12)
ggsave(plot=heritp2, device = "png", filename = "Plots/herit_met.png", dpi = "retina", height = 6, width = 12)
```

Cullis Heritability

```
traits <- c("PH", "NDF", "NDM", "OHSW", "PN", "GY")
library(sommer)
library(purrr)
library(dplyr)
calc_cullis <- function(tr){
  mod <- mmer(
    fixed = as.formula(
      paste0(tr, " ~ Env + Block:Env")
    ),
    random = ~ GEN + GEN:Env,
    rcov = ~ units,
    data = LB
  )
}
```

```

)
vg <- mod$sigma$GEN
C22 <- mod$PevU$GEN[[tr]]
n_g <- nrow(C22)
trC22 <- sum(diag(C22))

av2 <- 2/n_g *
  (trC22 -
    (sum(C22)-trC22)/(n_g-1))

H2 <- 1 - av2/(2*vg)

tibble(
  Trait = tr,
  Cullis_H2 = as.numeric(H2))
}
Cullis_all <- map_dfr(
  traits,
  calc_cullis)

```

iteration	LogLik	wall	cpu(sec)	restrained
1	-60.0291	14:49:31	0	0
2	-57.008	14:49:31	0	0
3	-56.1948	14:49:31	0	0
4	-56.1213	14:49:31	0	0
5	-56.1206	14:49:31	0	0

iteration	LogLik	wall	cpu(sec)	restrained
1	-38.7664	14:49:31	0	0
2	-26.3316	14:49:32	1	0
3	-23.5492	14:49:32	1	0
4	-23.3172	14:49:32	1	0
5	-23.3144	14:49:32	1	0
6	-23.3144	14:49:32	1	0

iteration	LogLik	wall	cpu(sec)	restrained
1	-80.8335	14:49:32	0	0
2	-80.7879	14:49:32	0	0
3	-80.7715	14:49:32	0	0
4	-80.7691	14:49:32	0	0
5	-80.769	14:49:32	0	0

iteration	LogLik	wall	cpu(sec)	restrained
1	-70.2583	14:49:32	0	0
2	-64.6976	14:49:32	0	0
3	-63.7078	14:49:32	0	0
4	-63.6435	14:49:32	0	0
5	-63.6432	14:49:32	0	0

iteration	LogLik	wall	cpu(sec)	restrained
1	-76.0017	14:49:33	1	0
2	-74.8831	14:49:33	1	0
3	-74.5778	14:49:33	1	0

4	-74.5462	14:49:33	1	0
5	-74.5456	14:49:33	1	0

iteration	LogLik	wall	cpu(sec)	restrained
1	-47.403	14:49:33	0	0
2	-46.9832	14:49:33	0	0
3	-46.8571	14:49:33	0	0
4	-46.8422	14:49:33	0	0
5	-46.8419	14:49:33	0	0

Cullis_all

```
# A tibble: 6 × 2
  Trait Cullis_H2
  <chr>      <dbl>
1 PH         0.528
2 NDF         0.752
3 NDM         0.403
4 OHSW        0.754
5 PN          0.281
6 GY          0.303
```

LRT

```
GYLRT = mixedmodel$GY$LRT
OHSWLRT = mixedmodel$OHSW$LRT
NDFLRT = mixedmodel$NDF$LRT
NDMLRT = mixedmodel$NDM$LRT
PHLRT = mixedmodel$PH$LRT
PNLRT = mixedmodel$PN$LRT
LRT = list(
  GY_LRT = GYLRT,
  OHSW_LRT = OHSWLRT,
  NDF_LRT = NDFLRT,
  NDM_LRT = NDMLRT,
  PH_LRT = PHLRT,
  PN_LRT = PNLRT)
LRT
```

```
$GY_LRT
      model npar  logLik    AIC    LRT Df Pr(>Chisq)
<none>      1   12 -2131.5 4286.9
(1 | GEN)    2   11 -2132.2 4286.5  1.5336  1  0.215571
(1 | GEN:ENV) 3   11 -2136.8 4295.6 10.6932  1  0.001075 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
$OHSW_LRT
      model npar  logLik    AIC    LRT Df Pr(>Chisq)
<none>      1   12 -823.50 1671.0
(1 | GEN)    2   11 -836.50 1695.0 26.008  1  3.4e-07 ***
(1 | GEN:ENV) 3   11 -825.72 1673.4  4.441  1  0.03509 *
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

\$NDF_LRT

	model	npar	logLik	AIC	LRT	Df	Pr(>Chisq)
<none>	1	12	-974.03	1972.1			
(1 GEN)	2	11	-984.81	1991.6	21.559	1	3.432e-06 ***
(1 GEN:ENV)	3	11	-983.98	1990.0	19.900	1	8.159e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

\$NDM_LRT

	model	npar	logLik	AIC	LRT	Df	Pr(>Chisq)
<none>	1	12	-1148.4	2320.9			
(1 GEN)	2	11	-1150.0	2322.0	3.0888	1	0.07883 .
(1 GEN:ENV)	3	11	-1151.6	2325.2	6.3046	1	0.01204 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

\$PH_LRT

	model	npar	logLik	AIC	LRT	Df	Pr(>Chisq)
<none>	1	12	-943.11	1910.2			
(1 GEN)	2	11	-947.08	1916.2	7.9428	1	0.004828 **
(1 GEN:ENV)	3	11	-952.98	1928.0	19.7427	1	8.86e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

\$PN_LRT

	model	npar	logLik	AIC	LRT	Df	Pr(>Chisq)
<none>	1	12	-2009.8	4043.6			
(1 GEN)	2	11	-2010.4	4042.8	1.2403	1	0.2654
(1 GEN:ENV)	3	11	-2018.0	4058.0	16.4317	1	5.044e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
write_xlsx(LRT, "LRT_LB.xlsx")
```

BLUPs

```
BLUPgen = list(
  GY_BLUP = mixedmodel$GY$BLUPgen,
  OHSW_BLUP = mixedmodel$OHSW$BLUPgen,
  NDF_BLUP = mixedmodel$NDF$BLUPgen,
  NDM_BLUP = mixedmodel$NDM$BLUPgen,
  PH_BLUP = mixedmodel$PH$BLUPgen,
  PN_BLUP = mixedmodel$PN$BLUPgen)
head(BLUPgen)
```

\$GY_BLUP

A tibble: 40 × 7

	Rank	GEN	Y	BLUPg	Predicted	LL	UL
<dbl>	<fct>	<dbl>	<dbl>		<dbl>	<dbl>	<dbl>
1	1	L06	1380.	164.	913.	612.	1214.
2	2	L32	1304.	144.	892.	591.	1193.

3	3	L18	1113.	122.	871.	570.	1172.
4	4	L24	1094.	115.	864.	563.	1165.
5	5	L30	1085.	112.	861.	560.	1162.
6	6	L07	1043.	96.8	846.	544.	1147.
7	7	L17	893.	96.7	845.	544.	1146.
8	8	L33	976.	72.7	821.	520.	1122.
9	9	L05	1020.	66.2	815.	514.	1116.
10	10	L23	923.	53.5	802.	501.	1103.

i 30 more rows

\$OHSW_BLUP

A tibble: 40 × 7

	Rank	GEN	Y	BLUPg	Predicted	LL	UL
	<dbl>	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	1	L40	38.6	5.34	37.5	34.3	40.7
2	2	L14	39.9	5.32	37.5	34.3	40.7
3	3	L25	37.2	4.18	36.4	33.2	39.6
4	4	L21	36.5	3.63	35.8	32.6	39.0
5	5	L26	36.1	3.30	35.5	32.3	38.7
6	6	L27	35.1	2.48	34.7	31.5	37.9
7	7	L23	34.9	2.38	34.6	31.4	37.8
8	8	L30	34.8	2.27	34.4	31.2	37.7
9	9	L18	34.3	1.87	34.0	30.8	37.3
10	10	L20	34.2	1.79	34.0	30.8	37.2

i 30 more rows

\$NDF_BLUP

A tibble: 40 × 7

	Rank	GEN	Y	BLUPg	Predicted	LL	UL
	<dbl>	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	1	L23	55.3	9.73	53.3	47.0	59.7
2	2	L28	55.3	9.73	53.3	47.0	59.7
3	3	L16	51.6	6.69	50.3	43.9	56.7
4	4	L26	51.2	6.42	50.0	43.7	56.4
5	5	L30	51.1	6.33	49.9	43.6	56.3
6	6	L19	50.8	6.06	49.7	43.3	56.0
7	7	L22	50.4	5.79	49.4	43.0	55.8
8	8	L27	50.3	5.70	49.3	43.0	55.7
9	9	L18	50.1	5.52	49.1	42.8	55.5
10	10	L40	49.9	5.34	49.0	42.6	55.3

i 30 more rows

\$NDM_BLUP

A tibble: 40 × 7

	Rank	GEN	Y	BLUPg	Predicted	LL	UL
	<dbl>	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	1	L11	101	6.41	94.2	85.2	103.
2	2	L40	101.	5.70	93.5	84.4	103.
3	3	L23	98.9	4.87	92.6	83.6	102.
4	4	L28	98.3	4.61	92.4	83.4	101.
5	5	L19	97.6	4.25	92.0	83.0	101.
6	6	L25	96.1	3.57	91.3	82.3	100.
7	7	L16	95.9	3.47	91.2	82.2	100.
8	8	L18	95.6	3.31	91.1	82.1	100.

```

9      9 L22      94.8  2.95      90.7  81.7  99.8
10     10 L38      91.2  2.77      90.5  81.5  99.6

```

```
# i 30 more rows
```

```
$PH_BLUP
```

```
# A tibble: 40 × 7
```

```

  Rank GEN      Y BLUPg Predicted    LL    UL
  <dbl> <fct> <dbl> <dbl>    <dbl> <dbl> <dbl>
1     1  1 L25     59.9  9.28     52.9  47.6  58.2
2     2  2 L28     52.0  4.53     48.1  42.8  53.4
3     3  3 L22     50.0  3.37     47.0  41.7  52.3
4     4  4 L30     48.5  2.43     46.0  40.7  51.3
5     5  5 L23     48.4  2.40     46.0  40.7  51.3
6     6  6 L27     48.2  2.26     45.9  40.6  51.1
7     7  7 L26     48.2  2.25     45.8  40.6  51.1
8     8  8 L18     47.5  1.88     45.5  40.2  50.8
9     9  9 L19     47.0  1.59     45.2  39.9  50.5
10    10 10 L17     47.5  1.57     45.2  39.9  50.5

```

```
# i 30 more rows
```

```
$PN_BLUP
```

```
# A tibble: 40 × 7
```

```

  Rank GEN      Y BLUPg Predicted    LL    UL
  <dbl> <fct> <dbl> <dbl>    <dbl> <dbl> <dbl>
1     1  1 L32     963 104.     679.  480.  878.
2     2  2 L05     847.  74.5     650.  451.  849.
3     3  3 L04     844.  73.6     649.  450.  848.
4     4  4 L02     832.  70.6     646.  447.  845.
5     5  5 L07     812.  69.7     645.  446.  844.
6     6  6 L33     803.  66.8     642.  443.  841.
7     7  7 L24     775.  57.4     633.  434.  831.
8     8  8 L17     744.  53.6     629.  430.  828.
9     9  9 L30     739.  45.3     620.  422.  819.
10    10 10 L18     734.  43.7     619.  420.  818.

```

```
# i 30 more rows
```

```
write_xlsx(BLUPgen, "BLUPgen_LB.xlsx")
```

```

sommer_models <- setNames(
  map(trait, \(tr)
    mmer(
      fixed = as.formula(
        paste0(tr, " ~ Env + Block:Env")
      ),
      random = ~ GEN + GEN:Env,
      rcov = ~ units,
      data = LB
    )
  ),
  trait)

```

```

iteration    LogLik      wall    cpu(sec)  restrained
1          -60.0291  14:49:34      0          0

```

2	-57.008	14:49:34	0	0
3	-56.1948	14:49:34	0	0
4	-56.1213	14:49:34	0	0
5	-56.1206	14:49:34	0	0

iteration	LogLik	wall	cpu(sec)	restrained
1	-38.7664	14:49:34	0	0
2	-26.3316	14:49:34	0	0
3	-23.5492	14:49:34	0	0
4	-23.3172	14:49:34	0	0
5	-23.3144	14:49:34	0	0
6	-23.3144	14:49:34	0	0

iteration	LogLik	wall	cpu(sec)	restrained
1	-80.8335	14:49:34	0	0
2	-80.7879	14:49:35	1	0
3	-80.7715	14:49:35	1	0
4	-80.7691	14:49:35	1	0
5	-80.769	14:49:35	1	0

iteration	LogLik	wall	cpu(sec)	restrained
1	-70.2583	14:49:35	0	0
2	-64.6976	14:49:35	0	0
3	-63.7078	14:49:35	0	0
4	-63.6435	14:49:35	0	0
5	-63.6432	14:49:35	0	0

iteration	LogLik	wall	cpu(sec)	restrained
1	-76.0017	14:49:35	0	0
2	-74.8831	14:49:35	0	0
3	-74.5778	14:49:35	0	0
4	-74.5462	14:49:35	0	0
5	-74.5456	14:49:35	0	0

iteration	LogLik	wall	cpu(sec)	restrained
1	-47.403	14:49:35	0	0
2	-46.9832	14:49:35	0	0
3	-46.8571	14:49:36	1	0
4	-46.8422	14:49:36	1	0
5	-46.8419	14:49:36	1	0

```
sommer_blups <- map(
  traits,
  function(tr){
    data.frame(
      GEN = gsub("GEN", "", names(sommer_models[[tr]]$U$GEN[[tr]])),
      BLUP_sommer = as.numeric(
        sommer_models[[tr]]$U$GEN[[tr]]
      )
    )
  }
)
```



```
)
names(sommer_blups) <- traits
```

```
#Compare BLUPs- gamem_met and mmer
library(dplyr)

compare_blups <- map_dfr(
  traits,
  function(tr){

    gamem_blup <-
      mixedmodel[[tr]]$BLUPgen |>
      dplyr::select(GEN, BLUPg)

    sommer_blup <- sommer_blups[[tr]]

    comp <- left_join(
      gamem_blup,
      sommer_blup,
      by = "GEN")
    tibble(
      Trait = tr,
      Correlation =
        cor(
          comp$BLUPg,
          comp$BLUP_sommer
        ),
      Max_Difference =
        max(
          abs(
            comp$BLUPg -
            comp$BLUP_sommer
          )
        )
    )

  }
)
compare_blups
```

```
# A tibble: 6 × 3
  Trait Correlation Max_Difference
<chr>      <dbl>          <dbl>
1 PH        1.000      0.0000106
2 NDF        1.000      0.000226
3 NDM        1.000      0.00106
4 OHSW       1.000      0.0000478
5 PN         1.000      0.101
6 GY         1.000      0.0933
```

```
library(dplyr)
library(tibble)
```

```
ebvMat <- data.frame(
  GEN = mixedmodel$GY$BLUPgen$GEN,
  GY = mixedmodel$GY$BLUPgen$Predicted,
  OHSW = mixedmodel$OHSW$BLUPgen$Predicted,
  NDF = mixedmodel$NDF$BLUPgen$Predicted,
  NDM = mixedmodel$NDM$BLUPgen$Predicted,
  PH = mixedmodel$PH$BLUPgen$Predicted,
  PN = mixedmodel$PN$BLUPgen$Predicted
) |>
column_to_rownames("GEN") |>
as.matrix()
dim(ebvMat)
```

```
[1] 40 6
```

```
head(ebvMat)
```

	GY	OHSW	NDF	NDM	PH	PN
L06	913.1034	37.51991	53.34655	94.18319	52.87625	678.7407
L32	892.2353	37.49419	53.34655	93.47780	48.12310	649.6380
L18	870.6745	36.36019	50.30658	92.64571	46.96580	648.8006
L24	863.9895	35.80589	50.03834	92.38568	46.02591	645.7856
L30	860.5489	35.47762	49.94893	92.02164	45.99476	644.8933
L07	845.5058	34.66053	49.68070	91.34557	45.85358	641.9544

```
save(ebvMat,file = "Data/ebvMat.rda")
```

```
trait_units <- c(
  GY = "kg/ha",
  OHSW = "g",
  PH = "cm",
  PN = "number",
  NDF = "number",
  NDM = "number")
plot_trait_heatmap <- function(x, trait_name) {
  gen_order <- sprintf("%02d", 1:40)
  df <- x[[trait_name]]$BLUPgen %>%
    filter(GEN %in% gen_order) %>%
    dplyr::mutate(
      GEN = factor(GEN, levels = gen_order),
      Trait = trait_name
    )
  ggplot(df, aes(x = GEN, y = Trait, fill = Predicted)) +
    geom_tile() +
    scale_fill_gradientn(
      colours = c("orange", "#E37383", "purple"),
      name = bquote(hat(y)~("(~.(trait_units[[trait_name]]~")))) +
    labs(x = NULL, y = NULL) +
    theme_minimal(base_size = 14) +
    theme(
      axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1, face = "bold"),
      axis.text.y = element_text(angle = 90, face = "bold"),
```

```

    legend.position = "right",
    panel.grid = element_blank()
}
traits <- c("GY", "OHSW", "PH", "PN", "NDF", "NDM")
BLUP_heatmaps <- lapply(traits, function(trait) {
  plot_trait_heatmap(mixedmodel, trait)
})
library(ggpubr)
blups_heatmap_panel <- ggarrange(
  plotlist = BLUP_heatmaps,
  ncol = 2, nrow = 3,
  labels = c("A", "B", "C", "D", "E", "F"), font.label = list(size = 10, face = "bold"),
  align = "v")
ggsave(
  "BLUPs_LB.pdf",
  blups_heatmap_panel,
  width = 14,
  height = 12,
  dpi = 600)

```

```

trait_units <- c(
  GY  = "kg/ha",
  OHSW = "g",
  PH  = "cm",
  PN  = "number",
  NDF  = "number",
  NDM  = "number")
plot_trait_dot <- function(x, trait_name){
  gen_order <- sprintf("L%02d", 1:40)
  df <- x[[trait_name]]$BLUPgen %>%
    filter(GEN %in% gen_order) %>%
    dplyr::mutate(
      GEN = factor(GEN, levels = gen_order),
      Trait = trait_name)
  ggplot(df, aes(x = Trait, y = GEN)) +
    geom_point(
      aes(
        size = round(abs(Predicted)),
        color = Predicted),
      alpha = 0.75) +
    scale_color_gradientn(
      colours = c("orange", "lightpink", "purple"),
      name = bquote(hat(y)~("~.(trait_units[[trait_name]])~")),
      guide = guide_colorbar(
        barheight = unit(0.3, "cm"),
        barwidth = unit(4, "cm"))) +

    scale_size(range = c(8,11), guide = "none") +

    labs(
      x = NULL,
      y = NULL) +

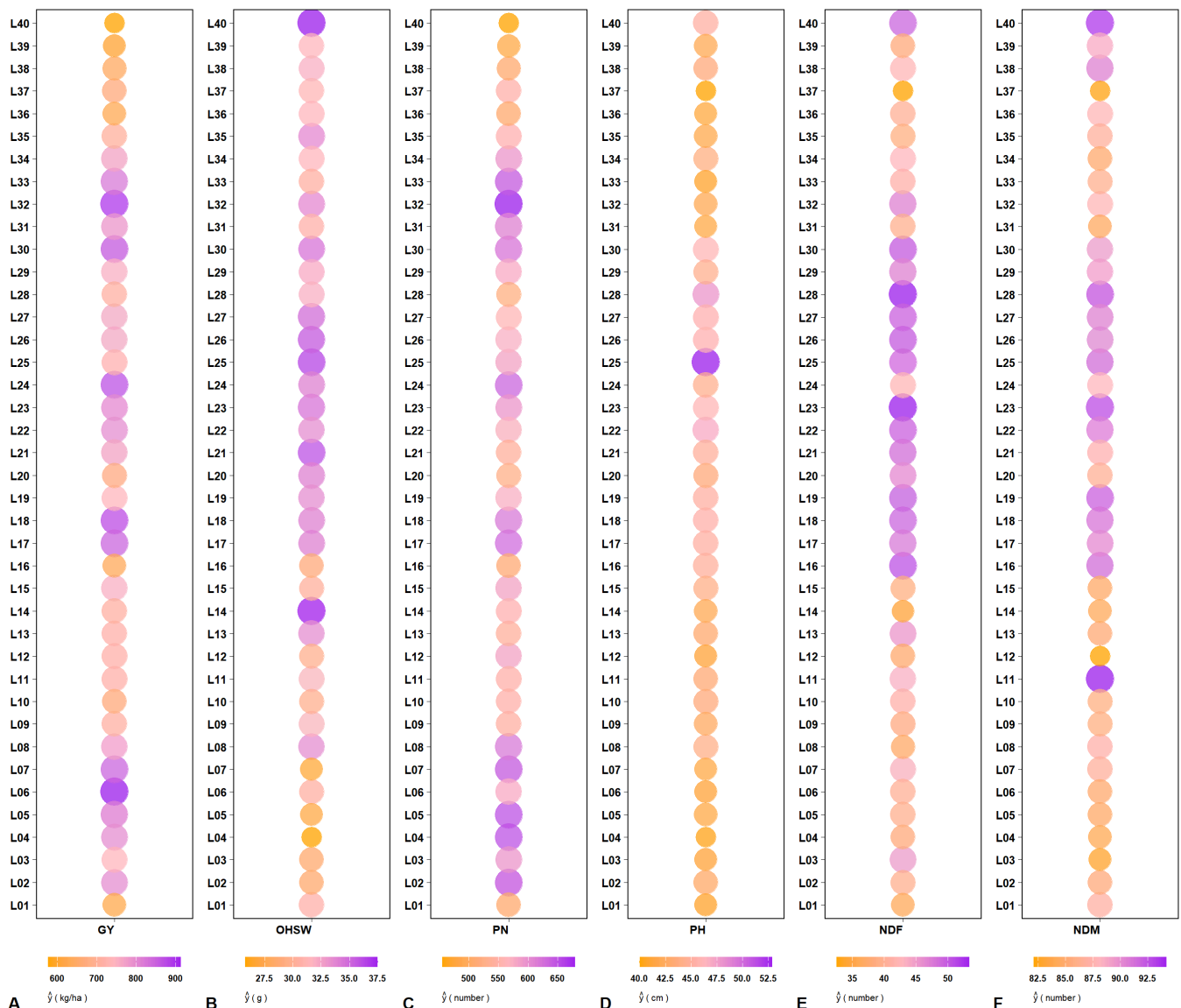
```

```
theme_linedraw(base_size = 12) +
theme(
  axis.text.x = element_text(
    angle = 0,
    hjust = 1,
    vjust = 1,
    face = "bold"),
  axis.text.y = element_text(face = "bold"),
  panel.grid = element_blank(),
  legend.position = "bottom", legend.title.position = "bottom",
  legend.text = element_text(size = 7.5, face = "bold"),
  legend.title = element_text(size = 8, face = "bold"))
}
traits <- c("GY", "OHSW", "PN", "PH", "NDF", "NDM")

BLUP_dots <- lapply(traits, function(tr){
  plot_trait_dot(mixedmodel, tr)
})

library(ggpubr)
#BLUP_dots
```

```
blups_d_panel <- ggarrange(
  plotlist = BLUP_dots,
  ncol = 6, nrow = 1,
  labels = c("A", "B", "C", "D", "E", "F"), vjust = 0.11,
  font.label = list(face = "bold"), label.y = 0.01,
  align = "v")
blups_d_panel
```



```
ggsave(
  "BLUPs_d2_LB.pdf",
  blups_d_panel, height = 11, width = 13,
  dpi = 600)
```

Correlation

```
corrplot = plot(
  corr_coef(LB, NDF, NDM, PH, PN, GY, OHSW),
  type = "upper",
  diag = FALSE,
  reorder = T,
  signif = c("stars", "pval"),
  show = c("all", "signif"),
  p_val = 0.05,
  caption = TRUE,
  digits.cor = 2,
  digits.pval = 3,
  col.low = "orange",
  col.mid = "white",
  col.high = "purple",
```

```

lab.x.position = "top",
lab.y.position = "left",
legend.position = "right",
legend.title = "Pearson's\nCorrelation",
size.text.cor = 5,
size.text.signif = 5,
size.text.lab = 14)

```

```
ggsave("Correlation.pdf",corrplot, width = 7, height = 6, dpi = 600)
```

```

library(circlize)
# Correlation matrix
corr_mat <- matrix(
  c(
    1.000, 0.587, 0.205, -0.230, -0.264, -0.129,
    0.587, 1.000, 0.224, -0.214, -0.262, -0.131,
    0.205, 0.224, 1.000, 0.160, 0.177, 0.413,
    -0.230, -0.214, 0.160, 1.000, 0.861, 0.123,
    -0.264, -0.262, 0.177, 0.861, 1.000, 0.267,
    -0.129, -0.131, 0.413, 0.123, 0.267, 1.000),
  nrow = 6,
  byrow = TRUE)
traits <- c(
  "NDF",
  "NDM",
  "PH",
  "PN",
  "GY",
  "OHSW")
rownames(corr_mat) <- traits
colnames(corr_mat) <- traits
# Convert matrix to links
corr_df <- data.frame()
for(i in 1:(nrow(corr_mat)-1)){
  for(j in (i+1):ncol(corr_mat)){
    corr_df <- rbind(
      corr_df,
      data.frame(
        from = rownames(corr_mat)[i],
        to   = colnames(corr_mat)[j],
        value = corr_mat[i,j]))
  }
}
# Link colors
corr_df$link_col <- ifelse(
  corr_df$value > 0,
  "purple",
  "orange")
# Trait colors
grid_cols <- c(
  NDF = "grey60",
  NDM = "grey40",
  PH  = "grey60",

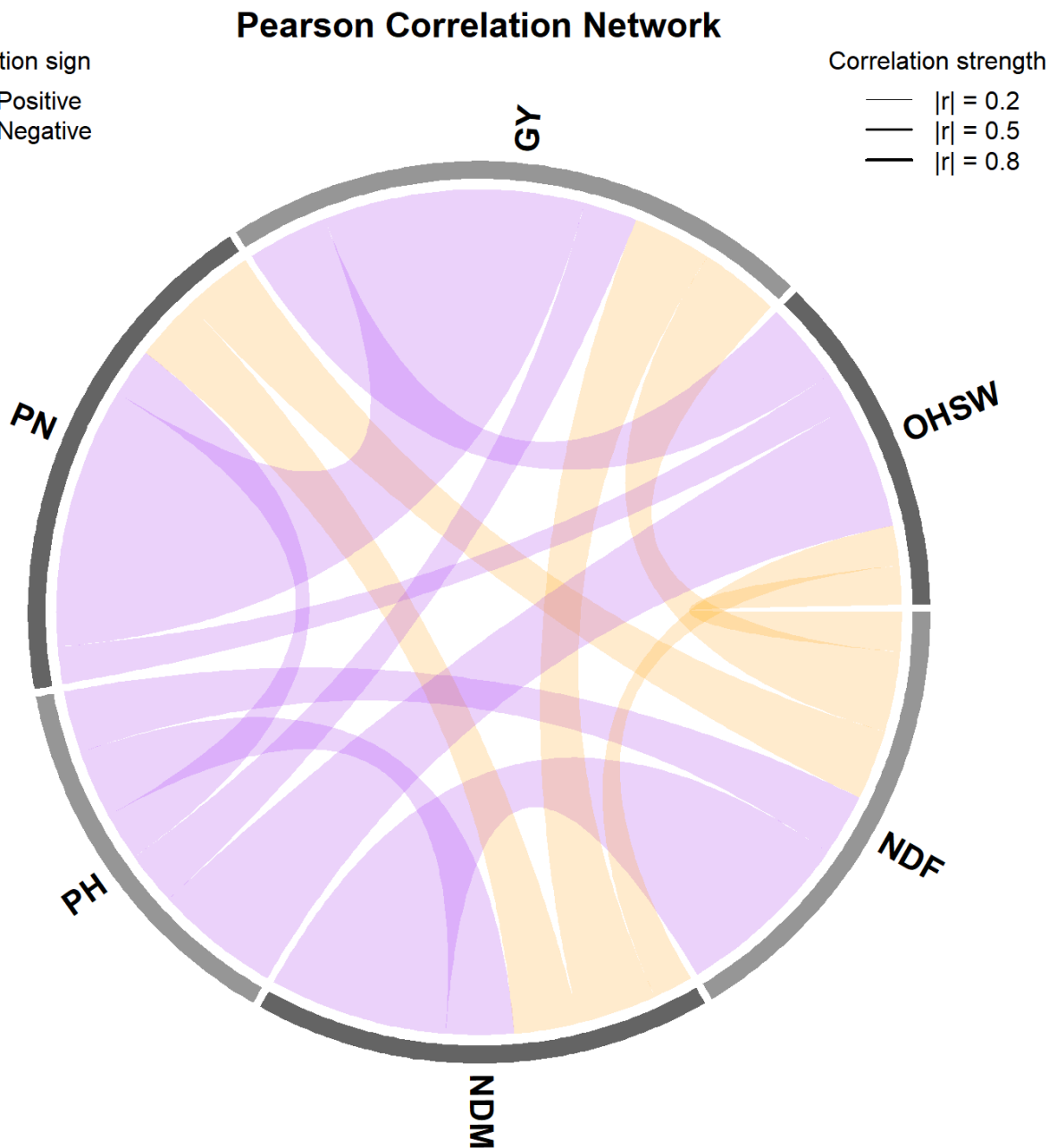
```

```

    PN   = "grey40",
    GY   = "grey60",
    OHSW = "grey40")
# Plot
circos.clear()
chordDiagram(
  x = corr_df[, c("from", "to", "value")],
  grid.col = grid_cols,
  col = corr_df$link_col,
  transparency = 0.80,
  directional = 0,
  # MUCH THINNER LINES
  link.lwd = 0.5 + abs(corr_df$value) * 2,
  annotationTrack = "grid",
  preAllocateTracks = list(
    track.height = 0.12))
# Trait labels
circos.track(
  track.index = 1,
  panel.fun = function(x,y){
    circos.text(
      CELL_META$xcenter,
      CELL_META$ylim[1],
      CELL_META$sector.index,
      facing = "clockwise",
      niceFacing = TRUE,
      adj = c(0,0.5),
      font = 2,
      cex = 1.3)
  },
  bg.border = NA
)
# Title
title(
  "Pearson Correlation Network",
  font.main = 2,
  cex.main = 1.4)
# Legends
legend(
  "topleft",
  legend = c(
    "Positive",
    "Negative"),
  col = c(
    "purple",
    "orange"),
  lwd = 4,
  title = "Correlation sign",
  bty = "n",
  cex = 1)
legend(
  "topright",
  legend = c(
    "|r| = 0.2",

```

```
"|r| = 0.5",  
"|r| = 0.8"),  
lwd = c(  
  0.5 + 0.2*2,  
  0.5 + 0.5*2,  
  0.5 + 0.8*2),  
col = "black",  
title = "Correlation strength",  
bty = "n",  
cex = 1)
```



```
pdf(  
  "Correlation_Chord_Diagram.pdf",  
  width = 8,  
  height = 8)  
dev.off()
```


GGE BIPLLOT

Which-Won-where

```

www <- list()

for(tr in traits){
  gge3_tr <- gge(LB, Env, GEN, resp = tr, svp = "symmetrical")

  p <- plot(gge3_tr,
            type = 3,
            col.env = "orange",
            col.gen = "purple",
            shape.gen = 21,
            shape.env = 23,
            col.alpha = 0.3,
            arrow.color = "purple",
            size.text.gen = 4,
            size.text.win = 8,
            size.text.env = 8,
            size.shape = 5,
            size.shape.win = 10,
            large_label = 10,
            repel = TRUE,
            max_overlaps = 40,
            repulsion = 8,
            col.line = "orange",
            plot_theme = theme_metan_minimal())

  p_try <- try({
    p2 <- p + labs(title = tr, subtitle = NULL) +
      theme(plot.title = element_text(hjust = 0.5, face = "bold"))
    p <- p2
    NULL
  }, silent = TRUE)

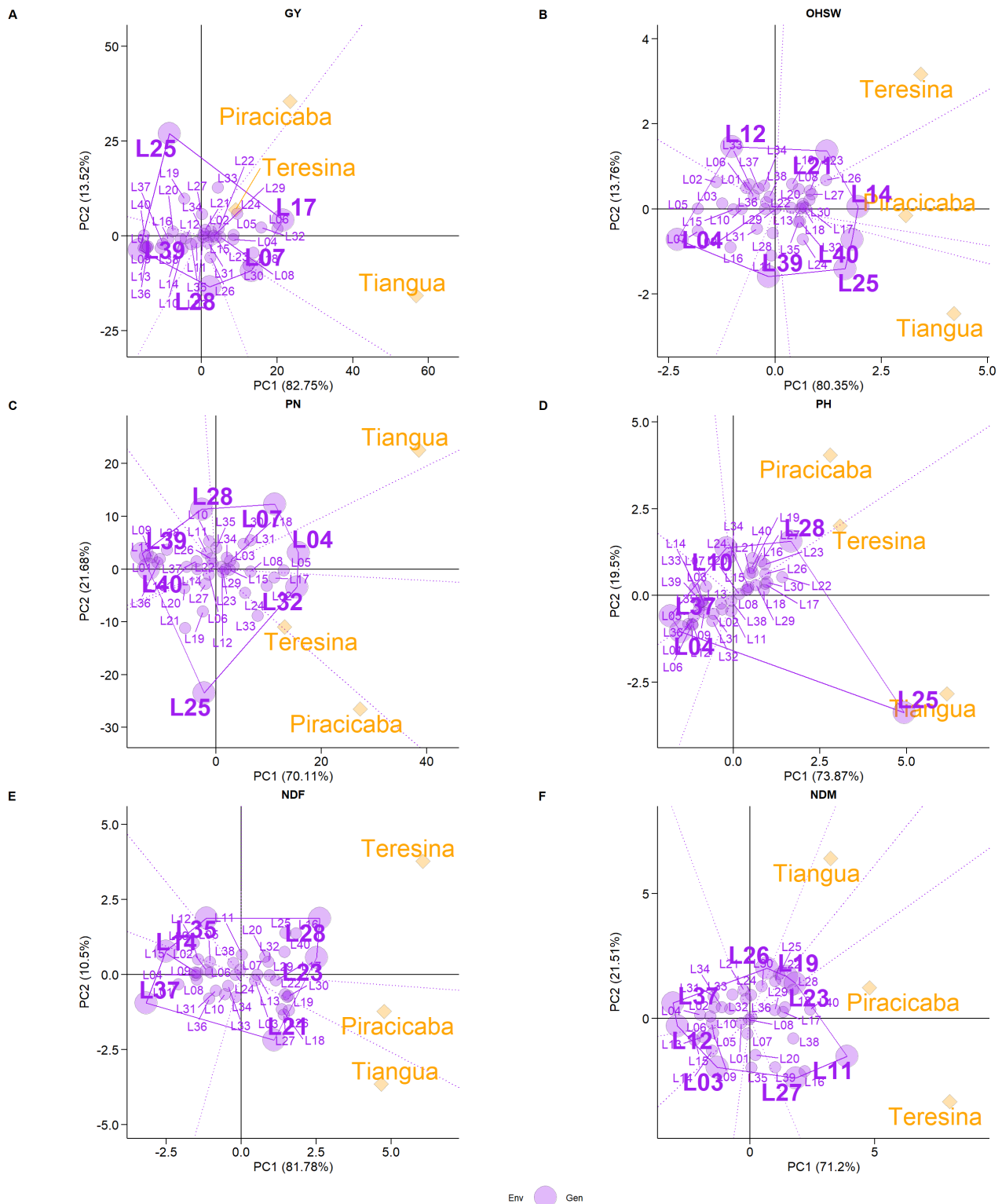
  if(inherits(p_try, "try-error") || !inherits(p, "ggplot")) {
    p_grob <- ggpubr::as_ggplot(p)
    p <- p_grob + labs(title = tr, subtitle = NULL) +
      theme(plot.title = element_text(hjust = 0.5, face = "bold"))
  }

  www[[tr]] <- p
}

www_lb <- ggarrange(
  www[["GY"]], www[["OHSW"]], www[["PN"]],
  www[["PH"]], www[["NDF"]], www[["NDM"]],
  ncol = 2, nrow = 3,
  common.legend = TRUE, legend = "bottom",
  labels = c("A", "B", "C", "D", "E", "F"),
  font.label = list(size = 12, face = "bold"),
  align = "hv",
  hjust = -0.5,

```

```
vjust = 1.5)
www_lb
```



```
ggsave("Which-won-Where2.pdf", www_lb, width = 14, height = 16, dpi = 600)
```

Mean_performance vs. stability

```
dir.create("Mean_performance_vs_stability", showWarnings = FALSE)

mps_list <- list()
```

```

for(tr in traits){
  gge2_tr <- gge(
    LB,
    Env,
    GEN,
    resp = tr,
    svp = "genotype")
# Plot type 2
p <- plot(gge2_tr,
  type = 2,
  col.env = "orange",
  col.gen = "purple",
  shape.gen = 21,
  shape.env = 23,
  col.alpha = 0.3,
  arrow.color = "purple",
  size.text.gen = 4,
  size.text.env = 5,
  size.shape = 5,
  large_label = 1,
  repel = TRUE,
  max_overlaps = 10,
  repulsion = 40,
  col.line = "orange",
  axis_expand = 1.2,
  plot_theme = theme_metan_minimal())

p_try <- try({
  p2 <- p + labs(title = tr, subtitle = NULL) +
    theme(plot.title = element_text(hjust = 0.5, face = "bold"))
  p <- p2
  NULL
}, silent = TRUE)

if(inherits(p_try, "try-error") || !inherits(p, "ggplot")) {
  p_grob <- ggpubr::as_ggplot(p)
  p <- p_grob + labs(title = tr, subtitle = NULL) +
    theme(plot.title = element_text(hjust = 0.5, face = "bold"))
}

mps_list[[tr]] <- p2}

mps_panel <- ggarrange(
  mps_list[["GY"]], mps_list[["OHSW"]], mps_list[["PN"]],
  mps_list[["PH"]], mps_list[["NDF"]], mps_list[["NDM"]],
  ncol = 2, nrow = 3,
  common.legend = TRUE, legend = "bottom",
  labels = c("A", "B", "C", "D", "E", "F"),
  font.label = list(size = 10, face = "bold"),
  align = "hv")
mps_panel = annotate_figure(
  mps_panel,
  top = text_grob("", size = 14),

```

```
fig.lab = "",  
fig.lab.pos = "bottom")
```

```
mps_GY <- gge(  
  LB,  
  Env,  
  GEN,  
  resp = "GY",  
  svp = "genotype")  
  
mpsp_GY <- plot(  
  mps_GY,  
  type = 2,  
  col.env = "orange",  
  col.gen = "purple",  
  col.line = "grey50",  
  size.text.gen = 4,  
  size.text.env = 5,  
  repel = TRUE,  
  repulsion = 0,  
  max_overlaps = 40,  
  axis_expand = 1.1,  
  plot_theme = theme_classic(base_size = 12)) +  
  labs(title = NULL, subtitle = NULL)  
  
mpsp_OHSW <- gge(  
  LB,  
  Env,  
  GEN,  
  resp = "OHSW",  
  svp = "genotype")  
  
mpsp_OHSW <- plot(  
  mpsp_OHSW,  
  type = 2,  
  col.env = "orange",  
  col.gen = "purple",  
  col.line = "grey50",  
  size.text.gen = 4,  
  size.text.env = 5,  
  repel = TRUE,  
  repulsion = 20,  
  max_overlaps = 30,  
  axis_expand = 1.08,  
  plot_theme = theme_classic(base_size = 14)) +  
  labs(title = NULL, subtitle = NULL)  
  
mpsp_PN <- gge(  
  LB,  
  Env,  
  GEN,  
  resp = "PN",  
  svp = "genotype")
```

```
mpspp_PN <- plot(  
  mpsp_PN,  
  type = 2,  
  col.env = "orange",  
  col.gen = "purple",  
  col.line = "grey50",  
  size.text.gen = 4,  
  size.text.env = 5,  
  repel = TRUE,  
  repulsion = 20,  
  max_overlaps = 30,  
  axis_expand = 1.1,  
  plot_theme = theme_classic(base_size = 14)) +  
  labs(title = NULL, subtitle = NULL)
```

```
mpsp_PH <- gge(  
  LB,  
  Env,  
  GEN,  
  resp = "PH",  
  svp = "genotype")
```

```
mpspp_PH <- plot(  
  mpsp_PH,  
  type = 2,  
  col.env = "orange",  
  col.gen = "purple",  
  col.line = "grey50",  
  size.text.gen = 4,  
  size.text.env = 5,  
  repel = TRUE,  
  repulsion = 20,  
  max_overlaps = 30,  
  axis_expand = 1.05,  
  plot_theme = theme_classic(base_size = 14)) +  
  labs(title = NULL, subtitle = NULL)
```

```
mpsp_NDF <- gge(  
  LB,  
  Env,  
  GEN,  
  resp = "NDF",  
  svp = "genotype")
```

```
mpspp_NDF <- plot(  
  mpsp_NDF,  
  type = 2,  
  col.env = "orange",  
  col.gen = "purple",  
  col.line = "grey50",  
  size.text.gen = 4,  
  size.text.env = 5,  
  repel = TRUE,
```

```
repulsion = 20,
max_overlaps = 30,
axis_expand = 1.1,
plot_theme = theme_classic(base_size = 14)) +
labs(title = NULL, subtitle = NULL)

mpsp_NDM <- gge(
  LB,
  Env,
  GEN,
  resp = "NDM",
  svp = "genotype")

mpspp_NDM <- plot(
  mpsp_NDM,
  type = 2,
  col.env = "orange",
  col.gen = "purple",
  col.line = "grey50",
  size.text.gen = 4,
  size.text.env = 5,
  repel = TRUE,
  repulsion = 20,
  max_overlaps = 30,
  axis_expand = 1.07,
  plot_theme = theme_classic(base_size = 14)) +
  labs(title = NULL, subtitle = NULL)

mpsp_GY <- mpsp_GY +
  ggtitle("GY") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))

mpspp_OHSW <- mpspp_OHSW +
  ggtitle("OHSW") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))

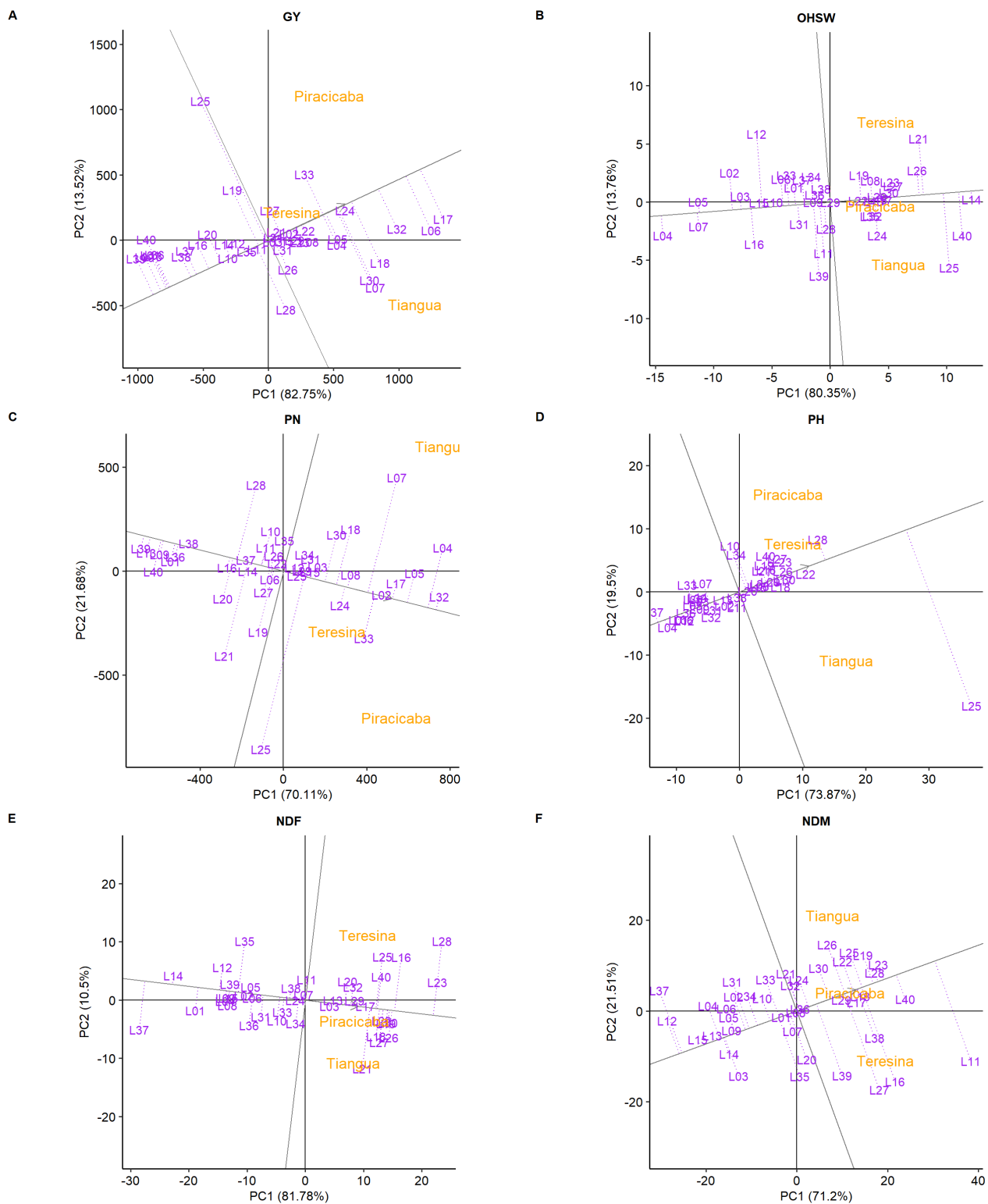
mpspp_PN <- mpspp_PN +
  ggtitle("PN") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))

mpspp_PH <- mpspp_PH +
  ggtitle("PH") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))

mpspp_NDF <- mpspp_NDF +
  ggtitle("NDF") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))

mpspp_NDM <- mpspp_NDM +
  ggtitle("NDM") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
```

```
common_theme <- theme(  
  legend.position = "bottom",  
  panel.grid = element_blank(),  
  plot.title = element_text(  
    hjust = 0.5,  
    face = "bold",  
    size = 12  
  )  
)  
mpsp_GY      <- mpsp_GY + common_theme  
mpspp_OHSW   <- mpspp_OHSW + common_theme  
mpspp_PN     <- mpspp_PN + common_theme  
mpspp_PH     <- mpspp_PH + common_theme  
mpspp_NDF    <- mpspp_NDF + common_theme  
mpspp_NDM    <- mpspp_NDM + common_theme  
mps_panel <- ggarrange(  
  mpsp_GY,  
  mpspp_OHSW,  
  mpspp_PN,  
  mpspp_PH,  
  mpspp_NDF,  
  mpspp_NDM,  
  ncol = 2,  
  nrow = 3,  
  common.legend = TRUE,  
  legend = "bottom",  
  labels = c("A", "B", "C", "D", "E", "F"),  
  font.label = list(  
    size = 12,  
    face = "bold"  
  )  
)  
mps_panel
```



```
ggsave("Mean Performance vs Stability_LB2.pdf", mps_panel, width = 14, height = 16, dpi = 600)
```

Discriminativeness vs. representativeness

```
library(ggplot2)
library(ggpubr)

dir.create("Discriminativeness_vs_Representativeness", showWarnings = FALSE)

dr_list <- list()
```



```

for(tr in traits){

  gge4_tr <- gge(
    LB,
    Env,
    GEN,
    resp = tr,
    centering = "environment",
    scaling = "none",
    svp = "environment"
  )

  p4 <- plot(gge4_tr,
    type = 4,
    col.env = "orange",
    col.gen = "purple",
    size.text.gen = 4,
    size.text.env = 4 ,
    repel = TRUE,
    repulsion = 10,
    col.line = "orange",
    size.line = 0.5,
    col.circle = "grey",
    col.alpha = 0.4,
    max_overlaps = 20,
    plot_theme = theme_metan_minimal())
  p_try <- try({
    p4_mod <- p4 +
      labs(title = tr, subtitle = NULL) +
      theme(plot.title = element_text(hjust = 0.5, face = "bold"))
    p4 <- p4_mod
    NULL
  }, silent = TRUE)
  if(inherits(p_try, "try-error") || !inherits(p4, "ggplot")) {
    p_grob <- ggpubr::as_ggplot(p4)
    p4 <- p_grob +
      labs(title = tr, subtitle = NULL) +
      theme(plot.title = element_text(hjust = 0.5, face = "bold"))
  }
  dr_list[[tr]] <- p4
}

dr_panel <- ggarrange(
  dr_list[["GY"]], dr_list[["OHSW"]], dr_list[["PN"]],
  dr_list[["PH"]], dr_list[["NDF"]], dr_list[["NDM"]],
  ncol = 2, nrow = 3,
  common.legend = TRUE, legend = "bottom",
  labels = c("A", "B", "C", "D", "E", "F"),
  font.label = list(size = 10, face = "bold"),
  align = "hv",
  hjust = -0.5,
  vjust = 1.5
)
dr_panel = annotate_figure(
  dr_panel,

```

```
top = text_grob("", size = 14),  
fig.lab = "",  
fig.lab.pos = "bottom"  
)
```

```
dr_GY <- gge(  
  LB,  
  Env,  
  GEN,  
  resp = "GY",  
  centering = "environment",  
  scaling = "none",  
  svp = "environment"  
)  
  
drp_GY <- plot(  
  dr_GY,  
  type = 4,  
  col.env = "orange",  
  col.gen = "purple",  
  size.text.gen = 3,  
  size.text.env = 5,  
  col.alpha = 0.2,  
  repel = TRUE,  
  repulsion = 5,  
  max_overlaps = 20,  
  col.line = "darkorange",  
  plot_theme = theme_classic(base_size = 12)  
) +  
  labs(title = NULL, subtitle = NULL)  
  
dr_OHSW <- gge(  
  LB,  
  Env,  
  GEN,  
  resp = "OHSW",  
  centering = "environment",  
  scaling = "none",  
  svp = "environment"  
)  
  
drp_OHSW <- plot(  
  dr_OHSW,  
  type = 4,  
  col.env = "orange",  
  col.gen = "purple",  
  size.text.gen = 3,  
  size.text.env = 5,  
  col.alpha = 0.2,  
  repel = TRUE,  
  repulsion = 5,  
  max_overlaps = 20,  
  col.line = "darkorange",
```

```
plot_theme = theme_classic(base_size = 12)
) +
  labs(title = NULL, subtitle = NULL)

dr_PN <- gge(
  LB,
  Env,
  GEN,
  resp = "PN",
  centering = "environment",
  scaling = "none",
  svp = "environment"
)

drp_PN <- plot(
  dr_PN,
  type = 4,
  col.env = "orange",
  col.gen = "purple",
  size.text.gen = 3,
  size.text.env = 5,
  col.alpha = 0.2,
  repel = TRUE,
  repulsion = 5,
  max_overlaps = 10,
  col.line = "darkorange",
  plot_theme = theme_classic(base_size = 12)
) +
  labs(title = NULL, subtitle = NULL)

dr_PH <- gge(
  LB,
  Env,
  GEN,
  resp = "PH",
  centering = "environment",
  scaling = "none",
  svp = "environment"
)

drp_PH <- plot(
  dr_PH,
  type = 4,
  col.env = "orange",
  col.gen = "purple",
  size.text.gen = 3,
  size.text.env = 5,
  col.alpha = 0.2,
  repel = TRUE,
  repulsion = 5,
  max_overlaps = 20,
  col.line = "darkorange",
  plot_theme = theme_classic(base_size = 12)
) +
```

```
labs(title = NULL, subtitle = NULL)

dr_NDF<- gge(
  LB,
  Env,
  GEN,
  resp = "NDF",
  centering = "environment",
  scaling = "none",
  svp = "environment"
)

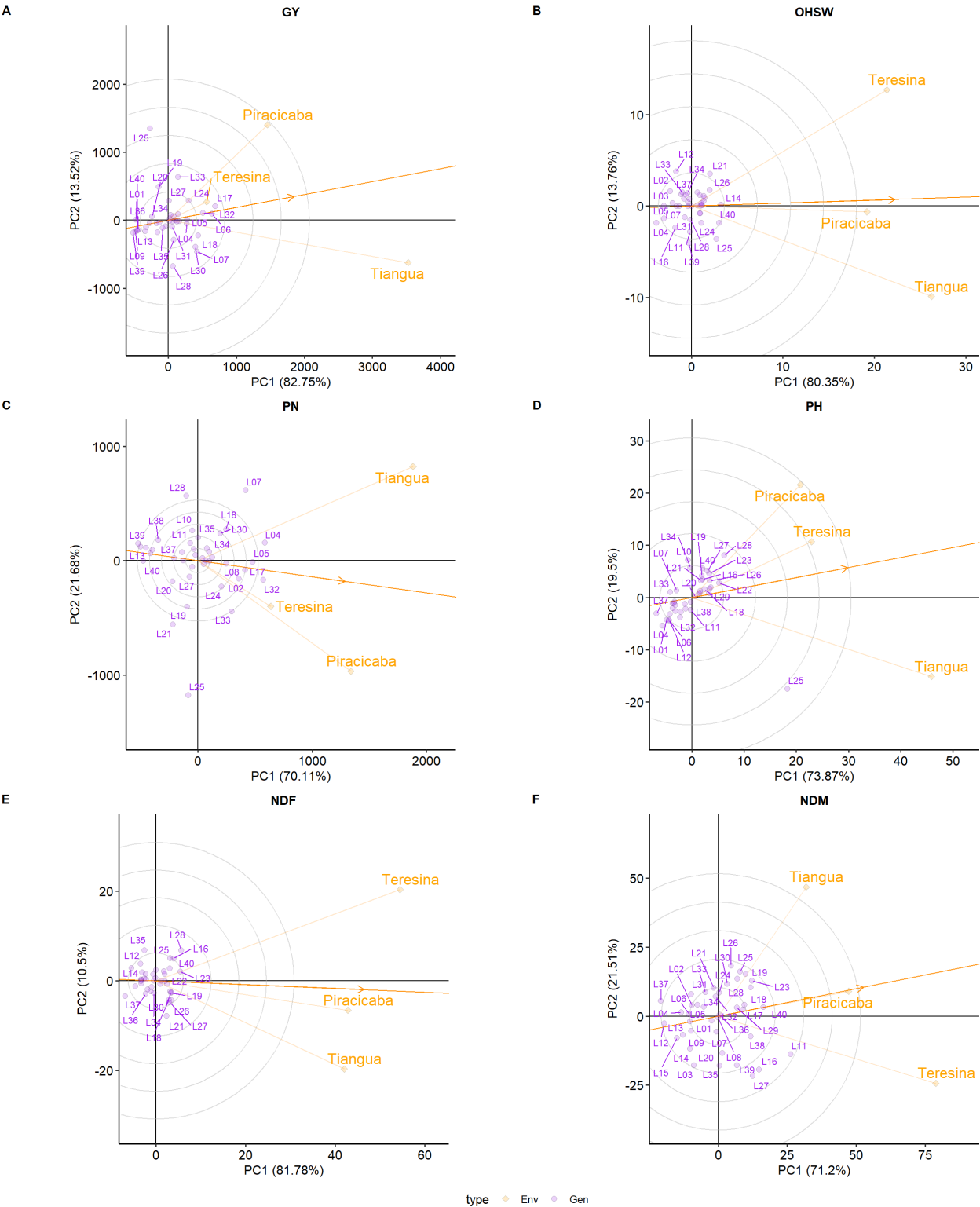
drp_NDF <- plot(
  dr_NDF,
  type = 4,
  col.env = "orange",
  col.gen = "purple",
  size.text.gen = 3,
  size.text.env = 5,
  col.alpha = 0.2,
  repel = TRUE,
  repulsion = 5,
  max_overlaps = 20,
  col.line = "darkorange",
  plot_theme = theme_classic(base_size = 12)
) +
  labs(title = NULL, subtitle = NULL)

dr_NDM<- gge(
  LB,
  Env,
  GEN,
  resp = "NDM",
  centering = "environment",
  scaling = "none",
  svp = "environment"
)

drp_NDM <- plot(
  dr_NDM,
  type = 4,
  col.env = "orange",
  col.gen = "purple",
  size.text.gen = 3,
  size.text.env = 5,
  col.alpha = 0.2,
  repel = TRUE,
  repulsion = 5,
  max_overlaps = 10,
  col.line = "darkorange",
  plot_theme = theme_classic(base_size = 12)
) +
  labs(title = NULL, subtitle = NULL)
```

```
drp_GY <- drp_GY +  
  ggtitle("GY") +  
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))  
  
drp_OHSW <- drp_OHSW +  
  ggtitle("OHSW") +  
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))  
  
drp_PN <- drp_PN +  
  ggtitle("PN") +  
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))  
  
drp_PH <- drp_PH +  
  ggtitle("PH") +  
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))  
  
drp_NDF <- drp_NDF +  
  ggtitle("NDF") +  
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))  
  
drp_NDM <- drp_NDM +  
  ggtitle("NDM") +  
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))  
common_theme <- theme(  
  legend.position = "bottom",  
  panel.grid = element_blank(),  
  plot.title = element_text(  
    hjust = 0.5,  
    face = "bold",  
    size = 12))  
drp_GY <- drp_GY + common_theme  
drp_OHSW <- drp_OHSW + common_theme  
drp_PN <- drp_PN + common_theme  
drp_PH <- drp_PH + common_theme  
drp_NDF <- drp_NDF + common_theme  
drp_NDM <- drp_NDM + common_theme  
dr_panel <- ggarrange(  
  drp_GY,  
  drp_OHSW,  
  drp_PN,  
  drp_PH,  
  drp_NDF,  
  drp_NDM,  
  ncol = 2,  
  nrow = 3,  
  common.legend = TRUE,  
  legend = "bottom",  
  labels = c("A", "B", "C", "D", "E", "F"),  
  font.label = list(  
    size = 12,  
    face = "bold"  
  )  
)
```

```
)  
dr_panel
```



```
ggsave("Discriminativeness vs Representativeness_LB.pdf", dr_panel, width = 14, height = 16, d
```

Simultaneous Selection

Desire Gain Index

```
# DESIRE INDEX FUNCTION
gainDiff <- function(ebvMat, b_ebv, dGain, nSel){
  take <- order(ebvMat %*% b_ebv, decreasing = TRUE)[1:nSel]
  mu_Sel <- colMeans(ebvMat[take, ])
  resp <- mu_Sel - colMeans(ebvMat)
  resp <- resp / sqrt(sum(resp^2))
  sum((resp - dGain)^2)
}
```

```
ebvMat <- ebvMat[complete.cases(ebvMat), ]
#COV MATRIX -EBVS
library(AlphaSimR)
```

Warning: pacote 'AlphaSimR' foi compilado no R versão 4.4.3

Carregando pacotes exigidos: R6

Anexando pacote: 'AlphaSimR'

O seguinte objeto é mascarado por 'package:ggpubr':

mutate

O seguinte objeto é mascarado por 'package:metan':

mutate

O seguinte objeto é mascarado por 'package:dplyr':

mutate

```
(G_ebv <- AlphaSimR::popVar(ebvMat))
```

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]
[1,]	6619.9044	218.241188	423.56486	252.987379	189.859236	4066.9162
[2,]	218.2412	7.357991	14.14453	8.392790	6.275343	134.1794
[3,]	423.5649	14.144528	28.50691	16.749936	12.432327	263.6634
[4,]	252.9874	8.392790	16.74994	10.010313	7.477405	156.8425
[5,]	189.8592	6.275343	12.43233	7.477405	6.242205	117.3492
[6,]	4066.9162	134.179444	263.66338	156.842455	117.349159	2535.0429

```
#desired gains
desiredGain <- c(
  GY = 2,
  OHSW = 1,
  NDF = -1,
  NDM = -1,
  PH = 0,
  PN = 1
)
```

```
desiredGain <- desiredGain / sqrt(sum(desiredGain^2))
sum(desiredGain^2)
```

```
[1] 1
```

```
b_ebv <- solve(G_ebv) %*% desiredGain
```

```
nSel <- 5
opt <- optim(
  par      = b_ebv,
  fn       = gainDiff,
  ebvMat    = ebvMat,
  dGain     = desiredGain,
  nSel      = nSel
)
(opt_weights <- opt$par)
```

```
      [,1]
[1,] -0.06221013
[2,]  2.68757780
[3,] -0.13848027
[4,] -1.84924576
[5,]  0.42431249
[6,]  0.06686236
```

```
DesireIndex <- ebvMat %*% opt_weights
DesireIndex <- data.frame(
  GEN = rownames(ebvMat),
  DesireIndex = as.numeric(DesireIndex)) |>
  arrange(desc(DesireIndex))
head(DesireIndex, 10)
```

```
      GEN DesireIndex
1  L20    -69.04102
2  L27    -69.27469
3  L21    -69.55347
4  L06    -69.70366
5  L13    -69.76704
6  L14    -69.98727
7  L11    -70.28663
8  L34    -70.42555
9  L25    -70.68409
10 L12    -70.71810
```

```
write.xlsx(DesireIndex, "Desireindex.xlsx")
```

```
(selected <- DesireIndex[1:5, ])
```

```
      GEN DesireIndex
1  L20    -69.04102
2  L27    -69.27469
```



```
3 L21    -69.55347
4 L06    -69.70366
5 L13    -69.76704
```

```
sel_mat <- ebvMat[selected$GEN, ]

expResp <- colMeans(sel_mat) - colMeans(ebvMat)
scaleExpResp <- expResp / sqrt(sum(expResp^2))

df_vect = cbind(
  Desired = desiredGain,
  Achieved = scaleExpResp)
```

```
library(dplyr)
df <- dplyr::arrange(DesireIndex, dplyr::desc(DesireIndex)) |>
  dplyr::mutate(
    GEN = factor(GEN, levels = GEN),
    Selected = ifelse(GEN %in% selected$GEN, "Selected", "Not selected"))
```

```
library(ggplot2)

desiredgainplot = ggplot(df, aes(x = GEN, y = DesireIndex, fill = Selected)) +
  geom_col(width = 0.9, color = "black", size = 0.2) +
  coord_polar() +
  scale_fill_manual(
    values = c(
      "Selected" = "#6A1B9A",
      "Not selected" = "orange")) +
  labs(face = "bold",
    fill = NULL) +
  theme_minimal(base_size = 14) +
  theme(
    axis.text.x = element_text(face = "bold"),
    axis.title = element_blank(),
    axis.text.y = element_blank(),
    axis.ticks = element_blank(),
    panel.grid = element_blank(),
    plot.title = element_text(face = "bold", hjust = 0.5),
    legend.position = "bottom")
```

```
ggsave("desiredgainplot_LB.pdf", plot = desiredgainplot, width = 10, height = 8, dpi = 600)
```

```
h2_df <- gmd(mixedmodel, "h2")
```

Class of the model: waasb

Variable extracted: h2

```
h2 <- h2_df$h2
names(h2) <- h2_df$VAR
traits_DI <- colnames(ebvMat)
```

```

h2_DI <- h2[traits_DI]
library(dplyr)

genetic_gain_table <- function(ebvMat, selected_gen, h2){

  traits <- colnames(ebvMat)

  res <- lapply(traits, function(tr){

    Xo <- mean(ebvMat[, tr], na.rm = TRUE)
    Xs <- mean(ebvMat[selected_gen, tr], na.rm = TRUE)

    SD <- Xs - Xo
    SDp <- (SD / Xo) * 100

    SG <- SD * h2[tr]
    SGp <- (SG / Xo) * 100

    tibble(
      Trait      = tr,
      `h²` = round(h2[tr], 2),
      Xo        = round(Xo, 2),
      Xs        = round(Xs, 2),
      SD        = round(SD, 2),
      `SD (%)`  = round(SDp, 2),
      SG        = round(SG, 2),
      `SG (%)`  = round(SGp, 2)
    )
  })

  bind_rows(res)
}

Table3_DesireIndex <- genetic_gain_table(
  ebvMat = ebvMat,
  selected_gen = selected$GEN,
  h2 = h2_DI)

```

```

h2_cullis <- Cullis_all$Cullis_H2
names(h2_cullis) <- Cullis_all$Trait
h2_cullis

```

	PH	NDF	NDM	OHSW	PN	GY
	0.5280800	0.7520623	0.4029824	0.7543363	0.2807144	0.3034326

```

Table3_DesireIndex_Cullis <- genetic_gain_table(
  ebvMat = ebvMat,
  selected_gen = selected$GEN,
  h2 = h2_cullis)
Table3_DesireIndex_Cullis

```

```

# A tibble: 6 × 8
  Trait `h²`   Xo   Xs   SD `SD (%)`   SG `SG (%)`
  <chr> <dbl> <dbl> <dbl> <dbl>   <dbl> <dbl>   <dbl>

```

1	GY	0.3	749.	766.	17.7	2.36	5.36	0.72
2	OHSW	0.75	32.2	33.2	1.01	3.12	0.76	2.36
3	NDF	0.75	43.6	44.5	0.91	2.08	0.68	1.56
4	NDM	0.4	87.8	88.2	0.39	0.44	0.16	0.18
5	PH	0.53	43.6	44.8	1.24	2.84	0.65	1.5
6	PN	0.28	575.	586.	11.1	1.94	3.13	0.54

```
Table_Broad <- genetic_gain_table(
  ebvMat,
  selected$GEN,
  h2_DI) %>%
  dplyr::rename(
    H2_Broad = `h²`,
    SG_Broad = SG,
    `SG_Broad (%)` = `SG (%)`)

Table_Cullis <- genetic_gain_table(
  ebvMat,
  selected$GEN,
  h2_cullis) %>%
  dplyr::rename(
    H2_Cullis = `h²`,
    SG_Cullis = SG,
    `SG_Cullis (%)` = `SG (%)`)

CompareGain <- Table_Broad %>%
  dplyr::select(Trait, H2_Broad, SG_Broad, `SG_Broad (%)`) %>%
  left_join(
    Table_Cullis %>%
      dplyr::select(Trait, H2_Cullis, SG_Cullis, `SG_Cullis (%)`),
    by = "Trait")
CompareGain
```

A tibble: 6 × 7

	Trait	H2_Broad	SG_Broad	`SG_Broad (%)`	H2_Cullis	SG_Cullis	`SG_Cullis (%)`
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	GY	0.36	6.36	0.85	0.3	5.36	0.72
2	OHSW	0.81	0.81	2.52	0.75	0.76	2.36
3	NDF	0.8	0.73	1.67	0.75	0.68	1.56
4	NDM	0.47	0.18	0.21	0.4	0.16	0.18
5	PH	0.6	0.74	1.69	0.53	0.65	1.5
6	PN	0.33	3.73	0.65	0.28	3.13	0.54

```
library(knitr)
```

Warning: pacote 'knitr' foi compilado no R versão 4.4.3

```
tableDesire =kable(Table3_DesireIndex)
```

```
write.xlsx(Table3_DesireIndex, "Table3_DesireIndex.xlsx", rowNames = FALSE)
```

Populations

```
library(tibble)
populations <- tribble(
  ~Population, ~Cross, ~Male_origin, ~Female_origin, ~Derived_lines,
  "H25", "G25236 × BGP UFPI 628",
  "Buenos Aires, Argentina (CIAT)",
  "Piauí, Brazil (BGP-UFPI)",
  "L01-L16",

  "H46", "UC HASKELL × BGP UFPI 728",
  "California, USA (UC Davis)",
  "Piauí, Brazil (BGP-UFPI)",
  "L17-L30",

  "H50", "BGP UFPI 728 × BGP UFPI 628",
  "Piauí, Brazil (BGP-UFPI)",
  "Piauí, Brazil (BGP-UFPI)",
  "L31-L34",

  "H81", "BGP UFPI 628 × UC 92",
  "Piauí, Brazil (BGP-UFPI)",
  "California, USA (UC Davis)",
  "L35-L37",

  "H94", "BGP UFPI 728 × UC 92",
  "Piauí, Brazil (BGP-UFPI)",
  "California, USA (UC Davis)",
  "L38-L40")
populations
```

A tibble: 5 × 5

	Population	Cross	Male_origin	Female_origin	Derived_lines
	<chr>	<chr>	<chr>	<chr>	<chr>
1	H25	G25236 × BGP UFPI 628	Buenos Air...	Piauí, Brazi...	L01-L16
2	H46	UC HASKELL × BGP UFPI 728	California...	Piauí, Brazi...	L17-L30
3	H50	BGP UFPI 728 × BGP UFPI 628	Piauí, Bra...	Piauí, Brazi...	L31-L34
4	H81	BGP UFPI 628 × UC 92	Piauí, Bra...	California, ...	L35-L37
5	H94	BGP UFPI 728 × UC 92	Piauí, Bra...	California, ...	L38-L40

```
library(dplyr)
library(tidyr)
library(stringr)
```

Warning: pacote 'stringr' foi compilado no R versão 4.4.3

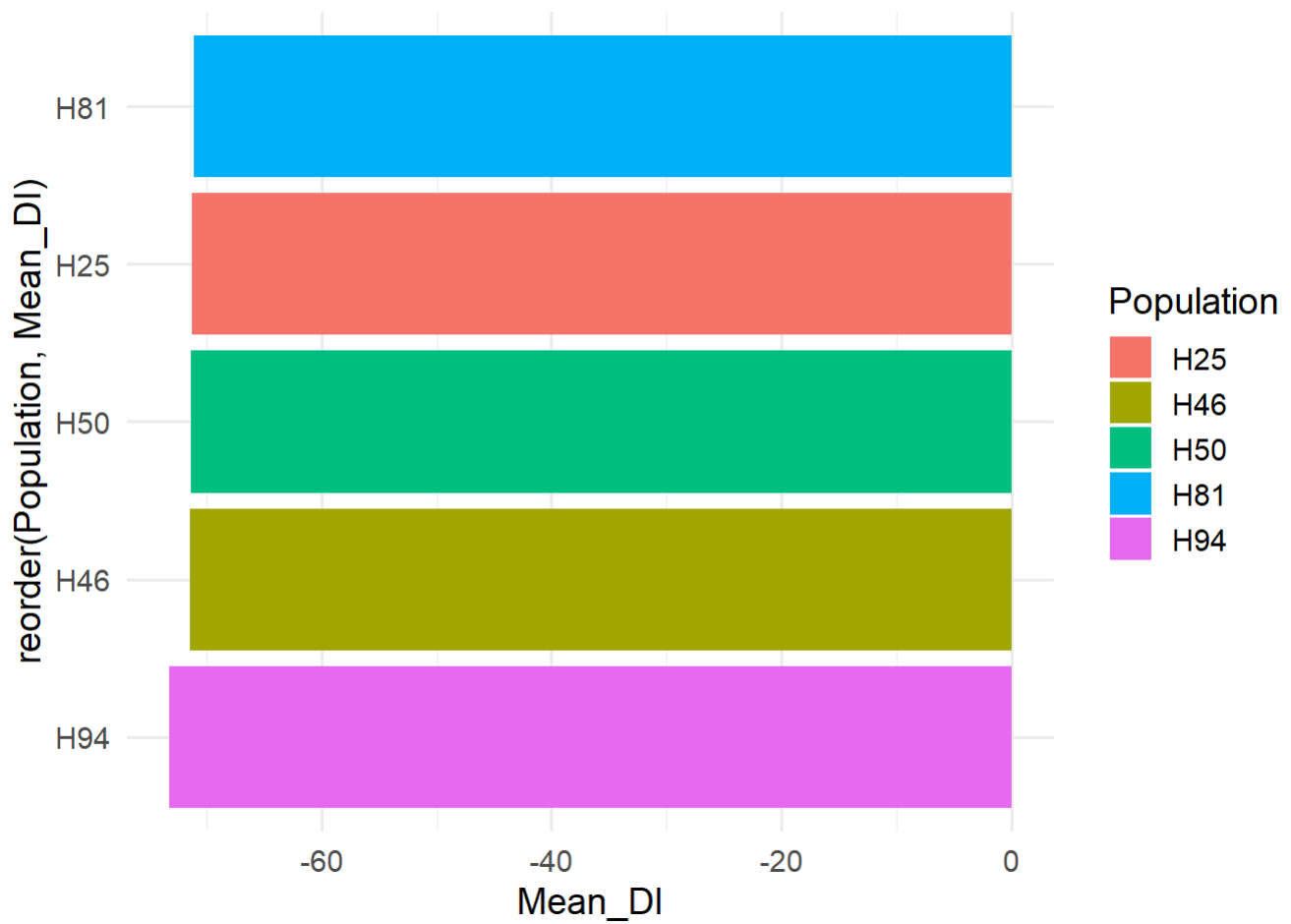
```
library(purrr)
pop_lines <- populations %>%
  rowwise() %>%
  dplyr::mutate(
    GEN = list(
      sprintf(
```

```
    "L%02d",
    seq(
      as.numeric(str_extract(Derived_lines, "(?<=L)\\d+")),
      as.numeric(str_extract(Derived_lines, "(?<=-L)\\d+"))
    )
  )
) %>%
unnest(GEN) %>%
ungroup()
```

```
DesireIndex_pop <- DesireIndex %>%
  left_join(
    pop_lines %>%
      dplyr::select(GEN, Population, Cross, Derived_lines),
    by = "GEN")
```

```
pop_summary <- DesireIndex_pop %>%
  group_by(Population, Cross) %>%
  summarise(
    Mean_DI = mean(DesireIndex),
    .groups = "drop")

ggplot(
  pop_summary,
  aes(
    x = reorder(Population, Mean_DI),
    y = Mean_DI,
    fill = Population)) +
  geom_col() +
  coord_flip() +
  theme_minimal(base_size = 14)
```



```
top5 <- selected$GEN
DesireIndex_pop <- DesireIndex_pop %>%
  dplyr::mutate(
    Elite = ifelse(GEN %in% top5, "Selected", "Not Selected")
  )
```

```
library(ggplot2)
library(ggrepel)

DGIPILOT = ggplot(
  DesireIndex_pop,
  aes(
    x = Population,
    y = DesireIndex
  )
) +

  geom_boxplot(
    aes(fill = Cross),
    width = 0.9,
    alpha = 0.13,
    linewidth = 0.5,
    outlier.shape = NA
  ) +

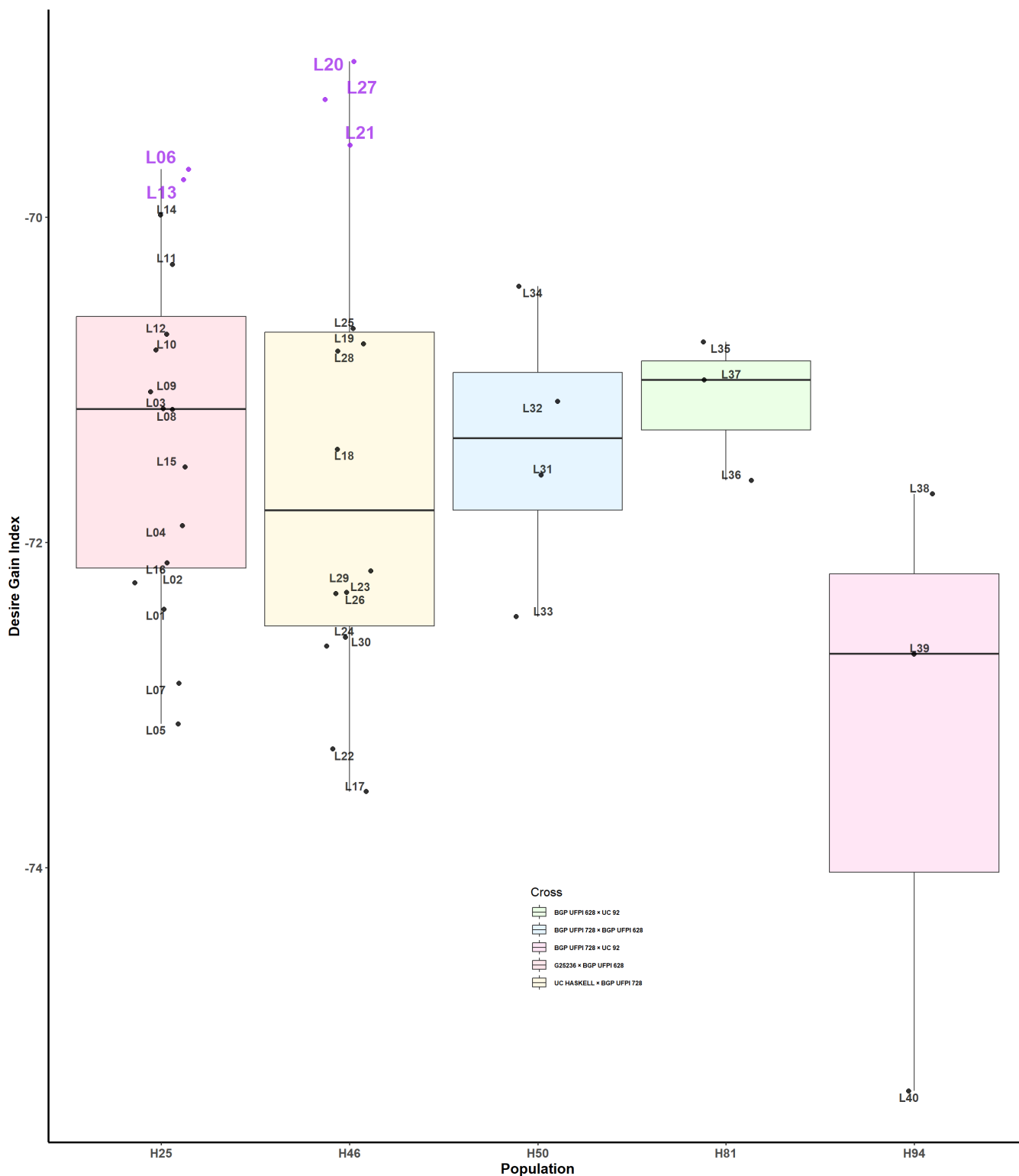
  # Points
  geom_jitter(
```

```

aes(
  color = Elite,
  shape = Elite),
width = 0.15,
size = 3,
alpha = 0.8) +
# Labels for non-selected lines
geom_text_repel(
  data = subset(DesireIndex_pop, Elite == "Not Selected"),
  aes(label = GEN),
  size = 4,
  fontface = "bold",
  alpha = 0.75,
  color = "black",
  max.overlaps = Inf,
  box.padding = 0,
  point.padding = 1,
  segment.alpha = 0) +
# Labels for selected lines
geom_text_repel(
  data = subset(DesireIndex_pop, Elite == "Selected"),
  aes(label = GEN),
  size = 6,
  fontface = "bold",
  alpha = 0.75,
  color = "purple",
  max.overlaps = Inf,
  box.padding = 0.3,
  point.padding = 1,
  segment.alpha = 0) +
scale_shape_manual(
  values = c(
    "Not Selected" = 20,
    "Selected" = 20
  ),
guide = "none") +
scale_color_manual(
  values = c(
    "Not Selected" = "black",
    "Selected" = "purple"),
guide = "none") +
scale_fill_manual(
values = c(
  "G25236 × BGP UFPI 628" = "#FF5E66",
  "UC HASKELL × BGP UFPI 728" = "#FFF75E",
  "BGP UFPI 728 × BGP UFPI 628" = "#5EB7FF",
  "BGP UFPI 628 × UC 92" = "#66FF5E",
  "BGP UFPI 728 × UC 92" = "#FF5EB7"
),
name = "Cross" )+
labs(
  x = "Population",
  y = "Desire Gain Index",
  color = NULL,

```

```
    shape = NULL) +  
theme_classic(base_size = 12) +  
theme(  
  panel.grid = element_blank(),  
  axis.line = element_line(  
    colour = "black",  
    linewidth = 1),  
  axis.title.x = element_text(  
    face = "bold",  
    size = 14),  
  axis.title.y = element_text(  
    face = "bold",  
    size = 14),  
  axis.text.x = element_text(  
    face = "bold",  
    size = 12),  
  axis.text.y = element_text(  
    face = "bold",  
    size = 12),  
  legend.text = element_text(  
    face = "bold",  
    size = 6),  
  legend.position = c(0.55, 0.18)  
)  
DGIPLLOT
```

Genotype by trait (GT) biplot

```
library(ggplot2)
gtb1 <- gtb(
  LB,
  GEN,
  resp = c(NDF, NDM, PH, PN, GY, OHSW))
gtb1plot <- plot(
  gtb1,
  col.env = "orange",
```

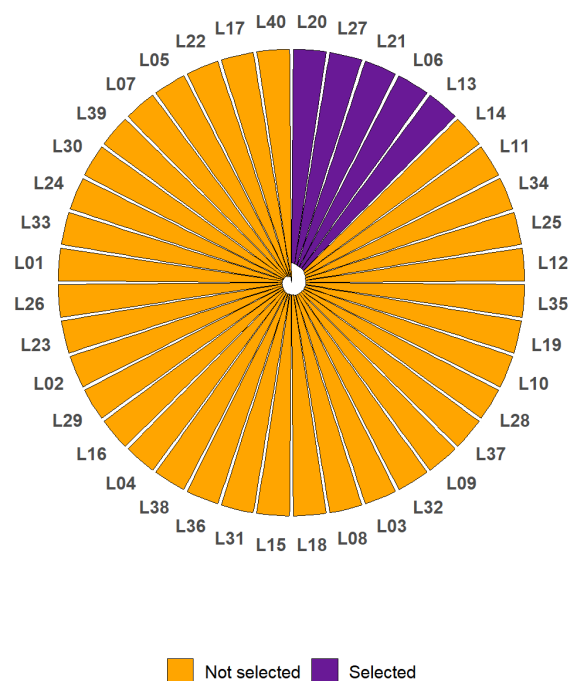
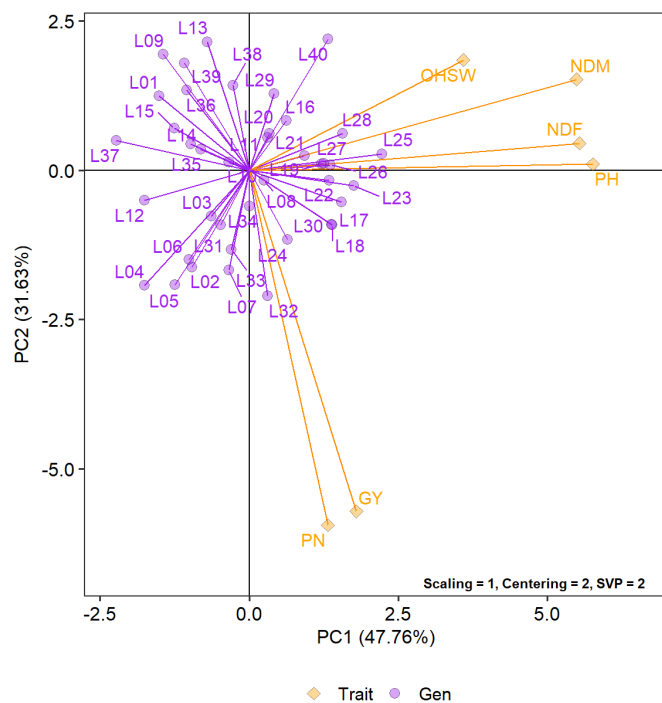
```
col.gen = "purple",
size.text.gen = 4,
size.text.env = 4,
repel = TRUE,
repulsion = 4,
col.line = "darkorange",
print.sub = FALSE,
plot_theme = theme_bw(base_size = 14) +
  theme(
    plot.title = element_blank(),
    plot.subtitle = element_blank(),
    panel.grid = element_blank(),
    legend.position = c(0.85, 0.2),
    legend.title = element_blank())
gtb1plot$layers[[5]]$aes_params$alpha <- 0.4
gtb1plot$layers[[5]]$aes_params$size <- 3
gtb1plot <- gtb1plot +
  annotate(
    "text",
    x = Inf,
    y = -Inf,
    label = "Scaling = 1, Centering = 2, SVP = 2",
    hjust = 1.1,
    vjust = -0.8,
    fontface = "bold",
    size = 3
  )
```

Arrange

```
gt_desired_panel <- ggarrange(gtb1plot, desiredgainplot,
  ncol = 2,
  common.legend = F, legend = "bottom",
  labels = c("A", "B"),
  font.label = list(size = 10, face = "bold"),
  align = "hv",
  hjust = -0.5,
  vjust = 1.5
)
gt_desired_panel
```

A

B



```
ggsave("gt_desired_panel_LB.pdf", plot = gt_desired_panel, dpi = 600)
```

Selection shift plot

```
library(dplyr)
library(tidyr)
library(ggplot2)
library(ggribbles)
trait_names <- c(
  GY_BLUP = "GY (kg/ha)",
  OHSW_BLUP = "OHSW (g)",
  PN_BLUP = "PN (Number)",
  PH_BLUP = "PH (cm)",
  NDF_BLUP = "NDF (days)",
  NDM_BLUP = "NDM (days)"
```

```
df_long <- bind_rows(
  lapply(names(BLUPgen), function(tr) {

    BLUPgen[[tr]] %>%
      dplyr::mutate(
        Trait = trait_names[[tr]],
        Status = ifelse(
          GEN %in% selected$GEN,
          "Selected",
          "Not Selected")) %>%
      dplyr::select(
        GEN,
        Trait,
        Status,
        Predicted)

  }))
```

```
library(ggplot2)
library(dplyr)
library()
df_long$Trait <- factor(
  df_long$Trait,
  levels = c(
    "GY (kg/ha)",
    "OHSW (g)",
    "PN (Number)",
    "PH (cm)",
    "NDF (days)",
    "NDM (days)"
  )
)
# Cullis heritability table
h2_df <- data.frame(
  Trait = c(
    "GY (kg/ha)",
    "OHSW (g)",
    "PN (Number)",
    "PH (cm)",
    "NDF (days)",
    "NDM (days)"
  ),
  H2 = c(
    0.303,
    0.754,
    0.281,
    0.528,
    0.752,
    0.403))

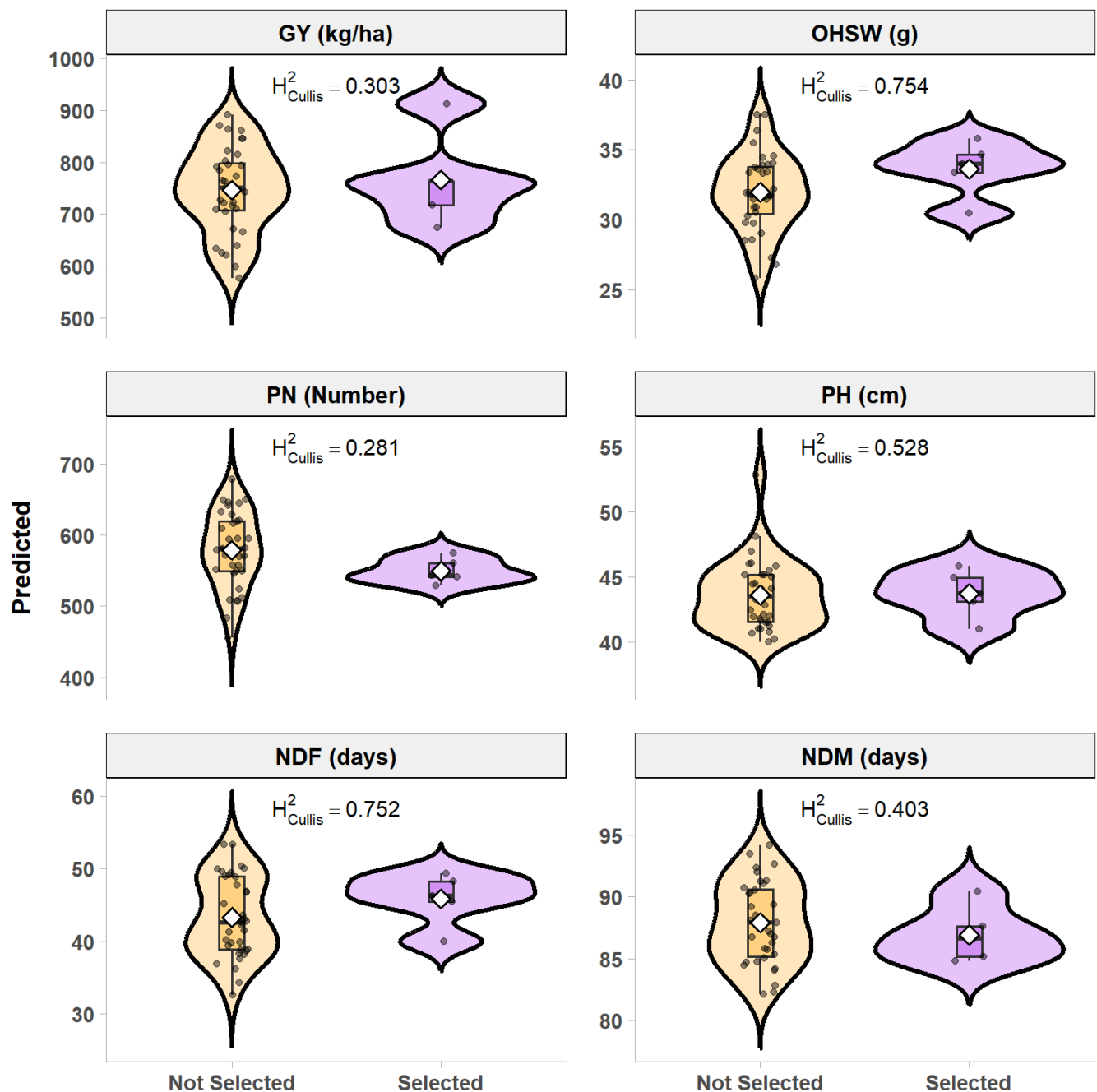
h2_df$Trait <- factor(
  h2_df$Trait,
  levels = levels(df_long$Trait))
# Plot
```

```

SelectionShiftPlot <- ggplot(
  df_long,
  aes(
    x = Status,
    y = Predicted,
    fill = Status)) +
  # Violin
  geom_violin(
    alpha = 0.25,
    color = "black",
    linewidth = 1.2,
    trim = FALSE) +
  # Boxplot
  geom_boxplot(
    width = 0.12,
    alpha = 0.3,
    linewidth = 0.6,
    outlier.shape = NA) +
  # Individual genotypes
  geom_jitter(
    width = 0.08,
    size = 1.5,
    alpha = 0.45,
    color = "black") +
  # Mean
  stat_summary(
    fun = mean,
    geom = "point",
    shape = 23,
    size = 3,
    fill = "white",
    color = "black",
    stroke = 1) +
  facet_wrap(
    ~Trait,
    scales = "free_y",
    ncol = 2, nrow = 3) +
  # Heritability annotation
  geom_text(
    data = h2_df,
    aes(
      x = 1.5,
      y = Inf,
      label = paste0(
        "H[Cullis]^2==",
        sprintf("%.3f", H2))),
    inherit.aes = FALSE,
    parse = TRUE,
    fontface = "bold",
    size = 4,
    vjust = 1.5) +
  scale_fill_manual(
    values = c(
      "Not Selected" = "orange",

```

```
    "Selected" = "purple")) +  
labs(  
  x = NULL,  
  y = "Predicted",  
  fill = NULL) +  
ggdist::theme_ggdist(base_size = 14) +  
theme(  
  panel.grid = element_blank(),  
  strip.background = element_rect(  
    fill = "grey95",  
    colour = "black",  
    linewidth = 0.5),  
  strip.text = element_text(  
    face = "bold",  
    size = 12),  
  axis.title.y = element_text(  
    face = "bold",  
    size = 13),  
  axis.text.x = element_text(  
    face = "bold",  
    size = 11),  
  axis.text.y = element_text(  
    face = "bold",  
    size = 11),  
  legend.position = "none",  
  legend.text = element_text(  
    face = "bold",  
    size = 11),  
  panel.spacing = unit(1.2, "lines"))  
SelectionShiftPlot
```



```
ggsave("SelectionShiftPlot.pdf", plot = SelectionShiftPlot,
       width = 8,
       height = 8,
       dpi = 600)
```

Density plot

```
ggsave(plot = densplot, "densplot.pdf", width = 14, height = 8, dpi = 600)
```

Environmental Similarity

```
library(EnvRtype)
library(terra)
library(raster)
```

```
library(dplyr)
library(ggplot2)
library(reshape2)

# ENVIRONMENTS
env.i <- c(
  "Piracicaba (SP)",
  "Teresina (PI)",
  "Tianguá (CE)"
)

lat <- c(-22.708333, -5.093889, -3.732222)
lon <- c(-47.636667, -42.784722, -41.012222)

plant.date <- c("2024-12-19", "2023-03-14", "2023-02-14")
harv.date <- c("2025-04-15", "2023-07-14", "2023-06-14")

# WEATHER DATA
df.clim <- get_weather(
  env.id = env.i,
  lat = lat,
  lon = lon,
  start.day = plant.date,
  end.day = harv.date
)

# CLIMATE COVARIATES
var.clim <- c("T2M", "VPD", "PRECTOT", "GWETTOP")

df.clim$DATE <- as.Date(as.character(df.clim$YYYYMMDD), format = "%Y%m%d")
df.clim$YEAR <- as.numeric(format(df.clim$DATE, "%Y"))
df.clim$MM <- as.numeric(format(df.clim$DATE, "%m"))
df.clim$DD <- as.numeric(format(df.clim$DATE, "%d"))

EC <- W_matrix(
  env.data = df.clim,
  env.id = "env",
  var.id = var.clim,
  statistic = "mean"
)

EC <- data.frame(env = rownames(EC), EC, row.names = NULL)

# SOIL RASTER
soilraster <- raster(
  "C:/Users/joaop/Documents/GitHub/Kenya Maps/KenyaMaps/Soil_Raster/hwsd_domi.tif"
)

coords <- data.frame(env = env.i, lon = lon, lat = lat)

soil_values <- extract(soilraster, coords[, c("lon", "lat")])

soil_class <- data.frame(env = env.i, soil_class = soil_values)
```



```

soil_class_names <- data.frame(
  soil_class = c(1, 10),
  class_name = c("Lixisol", "Regosol")
)

soil_class <- left_join(soil_class, soil_class_names, by = "soil_class")

# MERGE CLIMATE + SOIL
EC_soil <- left_join(
  EC,
  soil_class[, c("env", "class_name")],
  by = "env"
)

# DUMMY VARIABLES (FIXED)
soil_dummy <- model.matrix(~ class_name - 1, data = EC_soil)
soil_dummy <- as.data.frame(soil_dummy)

# FINAL MATRIX
EC_final <- cbind(
  EC_soil %>% dplyr::select(-class_name),
  soil_dummy
)

EC_numeric <- EC_final %>% dplyr::select(where(is.numeric))
EC_numeric <- as.matrix(EC_numeric)

rownames(EC_numeric) <- EC_final$env

# IMPUTE NA
for (i in 1:ncol(EC_numeric)) {
  EC_numeric[is.na(EC_numeric[, i]), i] <- mean(EC_numeric[, i], na.rm = TRUE)
}

# ENVIRONMENTAL KERNEL
K_E <- env_kernel(env.data = EC_numeric)[[2]]

# HEATMAP DATA
K_df <- melt(K_E)

colnames(K_df) <- c("Env1", "Env2", "Similarity")

K_df <- K_df %>%
  dplyr::mutate(
    Env1 = factor(Env1, levels = rownames(K_E)),
    Env2 = factor(Env2, levels = colnames(K_E))
  ) %>%
  filter(as.numeric(Env1) >= as.numeric(Env2))

# PLOT
envsim <- ggplot(K_df, aes(Env1, Env2, fill = Similarity)) +

  geom_tile(color = "white", linewidth = 0.5) +

```

```

scale_fill_gradientn(
  colors = c("purple", "#b2abd2", "#f7f7f7", "#fdb863", "orange"),
  name = "Environmental Similarity"
) +

coord_equal() +

theme_minimal(base_size = 16) +

theme(
  panel.grid = element_blank(),

  axis.text.x = element_text(face = "bold", size = 12),
  axis.text.y = element_text(face = "bold", size = 12),

  axis.title = element_blank(),

  plot.title = element_text(
    face = "bold",
    hjust = 0.5,
    size = 18
  ),

  legend.position = "bottom",

  legend.title = element_text(face = "bold", size = 12),
  legend.text = element_text(face = "bold", size = 10)
)

```

```

library(igraph)
library(ggraph)
library(ggplot2)
env_df <- data.frame(
  env = c(
    "Piracicaba",
    "Teresina",
    "Tianguá"
  ),
  x = c(0, 1, 0.5),
  y = c(0, 0, 0.9)
)

envsim2 = ggplot(env_df, aes(x, y)) +

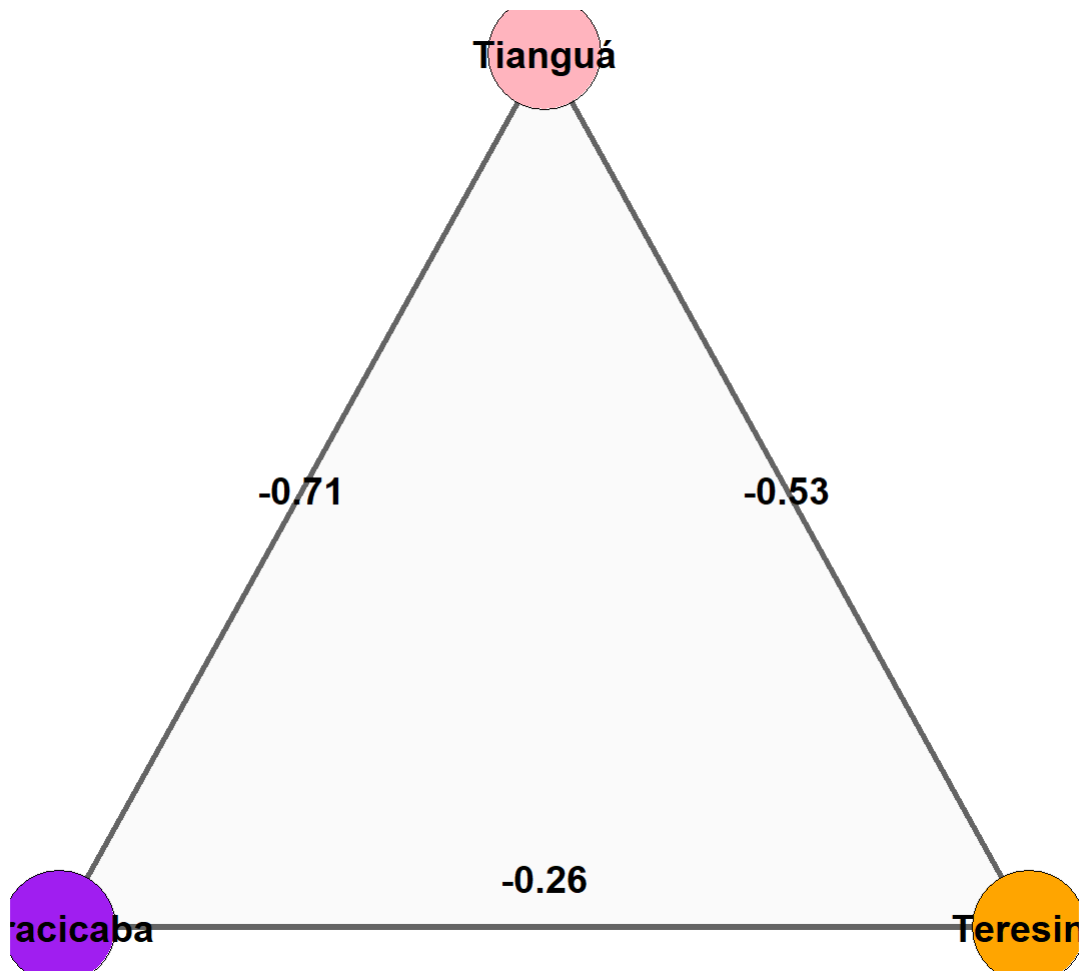
  # triangle
  geom_polygon(
    fill = "#F7F7F7",
    color = "grey40",
    linewidth = 1.2,
    alpha = 0.5
  ) +

  # environment points

```

```
geom_point(  
  aes(fill = env),  
  shape = 21,  
  size = 20,  
  color = "black",  
  stroke = 0  
) +  
  
# environment labels  
geom_text(  
  aes(label = env),  
  fontface = "bold",  
  size = 5,  
  nudge_y = 0,  
  nudge_x = 0  
) +  
  
# similarity values  
annotate(  
  "text",  
  x = 0.5,  
  y = 0.05,  
  label = round(K_E[1,2], 2),  
  size = 5,  
  fontface = "bold"  
) +  
  
annotate(  
  "text",  
  x = 0.25,  
  y = 0.45,  
  label = round(K_E[1,3], 2),  
  size = 5,  
  fontface = "bold"  
) +  
  
annotate(  
  "text",  
  x = 0.75,  
  y = 0.45,  
  label = round(K_E[2,3], 2),  
  size = 5,  
  fontface = "bold") +  
  
scale_fill_manual(  
  values = c(  
    "Piracicaba" = "purple",  
    "Teresina"   = "orange",  
    "Tianguá"    = "lightpink")) +  
  
coord_equal() +  
theme_void() +  
theme(  
  legend.position = "none"
```

```
)  
envsim2
```



```
ggsave("envsim2.pdf", plot = envsim2,  
  width = 8,  
  height = 8,  
  dpi = 600)
```